

Cooperation and Contract Design in Project Management with Outsourcing

Xiaoqiang Cai,^{a,b} Nicholas G. Hall,^c Siyi Wang,^{b,d} Feng Zhang^e

^aSchool of Data Science, The Chinese University of Hong Kong, Shenzhen 518172, China

^bShenzhen Research Institute of Big Data, Shenzhen 518172, China

xqcai@cuhk.edu.cn (✉)

^cDepartment of Operations & Business Analytics, Fisher College of Business, Ohio State University, Columbus, Ohio, USA

hall.33@osu.edu

^dSchool of Science and Engineering, The Chinese University of Hong Kong, Shenzhen 518172, China

siyiwang@link.cuhk.edu.cn

^eSuning Bank, Nanjing 210019, China

zhangfeng@suningbank.com

Abstract. We study a project management problem where the prime contractor needs to outsource tasks to subcontractors with the required resources. Successful execution of the project requires proper coordination among the subcontractors, as well as contract design by the prime contractor to incentivize the subcontractors. By modeling the subcontractors' coordination problem as a cooperative game, we develop a profit sharing scheme to facilitate the subcontractors' cooperation. We consider two contract designs for the prime contractor: a uniform contract across all subcontractors, and a nonuniform one that customizes incentives for each subcontractor. We propose efficient algorithms to solve the implicit optimization problems for optimal contract parameters. Computational experiments show that the pooling effect of subcontractors' cooperation mitigates the negative impact of poor estimates about the crashing cost and resource availability. We observe three unexpected results through the randomized computation experiments: (i) the subcontractors' profits may decrease if they provide false information; (ii) it is safer for the prime contractor to overestimate subcontractors' crashing costs than underestimate them; and (iii) uniform contracts deliver more project profit for the subcontractors in the coalitions.

Keywords: Project management with outsourcing, subcontracting, cooperative game, contract design

1. Introduction

30% of the world's economic activities, amounting to \$27 trillion annually, is organized as a projects (She et al. 2021). The use of project management as a methodology for planning and implementing business activities has become increasingly popular. One reason for this trend is newly developing applications, for example information technology, with substantial growth potential. A second reason is

shorter life cycles for products and services (Hall et al. 2016), which motivates the use of project management to bring new products and services to market more quickly. The third reason is the effectiveness of project management in implementing organizational change, and in responding to the increasingly frequent challenges of newer technologies, more intense competition, and more demanding and less predictable customers (1000ventures 2012,

Jalali et al. 2020).

Our work considers several issues in connection with the planning and scheduling of resource constrained projects. Overviews of the related research literature are provided by Herroelen et al. (1998) and Brucker et al. (1999). These works document the intractability of the resource constrained project scheduling problem, and discuss various approaches to solving it. Most of this literature considers the prime contractor or project owner as a single decision maker who owns all the necessary resources and controls all the tasks. For example, scheduling decisions that involve expediting, or *crashing*, the individual tasks are made by a sole decision maker to optimize some overall time and cost objective for the project. However, this perspective ignores the necessity of *outsourcing*, which is the procurement of products or services from external subcontractors. Since at least the early 1990's, firms have increasingly outsourced activities (Prahalad and Hamel 1990, Quinn and Hilmer 1994, Tully S 1994). The outsourced activities range from back-office business processes such as payroll processing and information technology (Garib 2009), to the manufacturing of components or entire products (Sturgeon 2002), to core activities related to new product development (Pisano and Verganti 2008). For example, revenue in the information technology outsourcing (ITO) segment is expected to reach US\$396.30 billion in 2022 with an annual growth rate (CAGR 2022-2027) of 8.20%¹. Qualified human resources (experts) are considered as one of the most important resources in ITO, and frequently required to be transferred among the different segments (Hedjazi

et al. 2019).

Further, in many projects, it is essential that specialized tasks are performed by outside subcontractors with necessary resources. The project manager or the prime contractor is mainly responsible for cost and schedule control. For instance, in construction projects, main contractors usually split a single project into several subprojects without knowing whether this decision is cost effective for subcontractors or not (Kumaraswamy and Matthews 2000, Perng et al. 2005, Asgari et al. 2014). Moreover, main contractors are more involved in site organization and management tasks, while subcontractors undertake up to 90% of the actual work of the total value of the construction project (Hartmann et al. 2009). Therefore, cooperation of subcontractors in a coalition may lead to less friction and increased profits in projects (Asgari et al. 2014, Moradi et al. 2019). Moradi et al. (2018) report one construction project in Iran, where the municipality of Tehran is the client of the prime contractor. Several subcontractors participate in the project with different phases and duration, since none of them has enough resources to accomplish the whole project at a specific time. Besides, those subcontractors are interested to form a coalition to reduce the activity time. Another example is agile global outsourced software development (AGOSD) in ITO, in which the client organizations outsource software development projects to external contractors, who are then using agile development practices within virtual project teams (subcontractors). In addition, the team members on this project come from all over the world, therefore collaboration

and efficient communication are of vital importance to the success of the entire project (Yu and Petter 2014, Dreesen et al. 2016). These realities necessitate the coordination of projects across organizational boundaries (Amaral et al. 2011), which is the topic of this paper.

In order to focus on the role of outsourcing in projects, we consider a project in which all the tasks are outsourced to a set of subcontractors by a prime contractor. All resources are owned by subcontractors, and the prime contractor is informed of the resource availability through subcontractors' tenders. The subcontractors can process their tasks independently. However, they can also cooperate with each other. Cooperation among subcontractors can motivate them to focus on project outcomes instead of individual outcomes (Matthews and Howell 2005). If they cooperate, then they share their resources and also coordinate their schedules to perform tasks, for mutual benefit. We model this problem as a cooperative game among the subcontractors. We further study the important issue of how the prime contractor should design contractual incentives to motivate the subcontractors to expedite their tasks appropriately. Usually, additional resources are required to accelerate the processing of a specific task, while compensation schemes including incentives and shared profits facilitate cooperation accordingly (Eriksson 2008). We model the contract design problem as a Stackelberg game, where the prime contractor is the leader of the game who specifies the contracts, and the subcontractors are followers who attempt to maximize their profits, given the contracts.

A cooperative coalition of subcontractors

can be formed from an existing network of available project subcontractors. Examples of such networks include *Global Projects Network* (2013) and *Global Quality & Services* (2013). In the projects, time interdependence and resource interdependence, such as joint objectives, joint project office, joint IT-database, team building activities, and dispute resolution techniques, directly facilitate trust and cooperation between subcontractors (Das and Teng 1998, Cheung et al. 2003). A survey (Smart Mobility Management 2013) identifies the main advantages of such networks as process optimization, improved communication and information sharing, and the solving of a particular customer problem. Hyun (1994) studies subcontractor partnerships in European automobile manufacturing, and concludes, "As the competitiveness of the company is not determined by the company itself, but by its supplier network, partnership commitments and effective supplier management become very important." Errasti et al. (2007) conduct an empirical study of subcontractor partnerships in the construction industry. They comment, "Generally, main subcontractors and subcontractors, operate within a project based system with the result that these organizations are increasingly developing horizontal and integrated organizational structures, which are geared toward the management of projects." Albino et al. (2007) discuss examples of horizontal cooperation to minimize unsatisfied demand. Perng et al. (2005) and Moradi et al. (2018) discuss the potential benefits of subcontractor cooperation in the construction industry, and how those benefits can be distributed among the participants

using concepts from cooperative game theory. [Chen and Hall \(2021\)](#) proposes incentive schemes that eliminate or mitigate Parkinson's Law in project management with the help of cooperative game theory.

We model the coordination problem of the subcontractors as a cooperative game ([Peleg and Sudhölter 2003](#)). Related works on general cooperative games include [Owen \(1975\)](#), [Samet and Zemel \(1984\)](#), [Granot \(1986\)](#). More closely related to our work are several previous studies that apply cooperative game concepts to project management. [Bergantiños and Sánchez \(2002\)](#) consider a project that experiences a delay in completion time. The identification of which tasks have caused the delay suggests how the cost of compensation to the client should be divided among the tasks. [Brânzei et al. \(2002\)](#) describe an approach based on an optimistic and a pessimistic game, and a second approach based on a serial cost sharing rule. The latter approach is extended to a weighted serial cost sharing rule by ([Castro et al. 2008](#)). Both the two previous studies assume that the project is inevitably delayed. [Estévez-Fernández et al. \(2005\)](#) study an expedited *project game* where tasks cannot be delayed. This game is shown to be convex. The authors also consider another project game where some tasks are delayed and other are expedited. Assuming that the delayable tasks are indeed delayed and the expeditable tasks outside the coalition are completed on time, this game is convex. [Castro et al. \(2007\)](#) also consider a project game, where the characteristic function for a given subset of activities represents the amount of expediting that these activities induce in the project, and show that

this game is balanced.

Whereas all the above studies consider a project that has already failed, our focus is on the performance of the project at the planning stage. Moreover, we address the more general situation where a payment is made at the completion of each task, rather than only at the completion of the entire project. Our model can therefore allow different tasks to have different per time unit incentives for expediting, and different per time unit penalties for late completion. This generalization is natural where there is an overall prime contractor and various subcontractors. It also allows for *milestone* payments in large scale projects. In projects that are expensive or of long duration, it is common for (sub)contractors to receive milestone payments. Practical examples occur most frequently within the construction and pharmaceutical industries. In the construction industry, the [U.S. Department of Transportation \(2013\)](#) summarizes this feature of projects, "Milestone payments or incentives are given to a contractor for completing a pre-determined milestone, on or ahead of schedule. Likewise, a disincentive can be incurred per day for failure to hit the target completion date." [Plaskoff \(2013\)](#) discusses how to set a construction and milestone payment schedule. In the pharmaceutical industry, examples of substantial milestone payments are provided by [San Diego Business Journal \(2013\)](#) for Isis Pharmaceuticals, and by the [Wall Street Journal \(2013\)](#) for Sucampo Corporation. Milestone payments are also well recognized and modeled within the academic literature ([Abu-Hiljeh and Ibbs 1989](#), [Shtub and Etgar 1997](#), [Vanhoucke et al. 2001](#)). We show that the more general game we

formulate, although not convex, is balanced, and develop a profit sharing scheme that ensures a stable cooperation within subcontractors. We also generalize these results to allow task lengthening, where a subcontractor can execute his task at a slower speed, in order to release resources to other subcontractors who can use them more effectively.

Our work is also related to the resource-constrained project scheduling problems in the context of time/cost tradeoffs. In the existing literature, the coalition of subcontractors is formed for profit-maximization and/or time-minimization (Asgari et al. 2014, Hafezalkotob et al. 2017, Moradi et al. 2019). Moradi et al. (2018) use the net present value of the project to evaluate the performance of different profit-sharing scheme in the grand coalition. However, we study the prime contractor's optimal contractual incentives to motivate the subcontractors to expedite their tasks, in order to minimize the completion cost associated with time. Li et al. (2021) study multi-project scheduling and resource allocation in the context of a decentralized time-cost tradeoff problem, and propose negotiation-based cooperative mechanism to resolve the problem accordingly. Teng et al. (2017) investigate the profit distribution scheme in a problem with multiple projects by using cooperative game theory. Hafezalkotob et al. (2017) consider cooperation and competition of subcontractors with interaction activities in a project with subcontracting. Their cooperative model is different from ours in several aspects, such as the objective of the joint action (to minimize the total time), the resource constraints (with only a budget constraint), and the decision variables of a coalition. Moreover,

they do not consider the contract design issue, which is much more complicated in the project management problem with subcontracting.

We study the contract design problem of the prime contractor in two cases: a uniform contract across all subcontractors, and a customized one for each subcontractor. We develop efficient algorithms to compute the optimal contract parameters in both cases, and evaluate the relative change in project performance achieved by a nonuniform contract. We computationally study the sensitivity of profit to the accuracy of parameter estimation by the prime contractor at the contract design stage, providing useful insights for the prime contractor's optimal contract design with outsourcing. Through computational experiments, we find that the pooling effect of the subcontractors' cooperation mitigates the impact of poor estimation about the crashing cost and resource availability. We also discover three unexpected results. First, when the subcontractors provide false information about their costs or resources, this may reduce the overall project profit and also their own profit. Second, it is safer for the prime contractor to overestimate the subcontractors' crashing costs than to underestimate them. Third, uniform contracts often deliver more overall project profit than nonuniform ones, although the prime contractor's share of profit is always larger under nonuniform contracts.

This paper is organized as follows. Section 2 defines our notation, and introduces the cooperative game among the subcontractors and the contractor design problem of the prime contractor. In Section 3, we show that the cooperative game of the subcontractors is balanced,

and identify a stable profit sharing scheme. Section 4 develops algorithms for the prime contractor to design optimal contracts with the subcontractors. Section 5 describes computational experiments to evaluate the sensitivity of parameter estimation. Finally, Section 6 provides a conclusion and some suggestions for future work.

2. Preliminaries

2.1 Problem Definition and Information Flow

The projects we consider contain a number of tasks, with precedence relations between them that can be characterized by an acyclic network. A firm, the *prime contractor* ("she"), has secured a contract with the project owner to perform the project. However, the prime contractor does not have resources for performing the tasks, and hence the tasks are outsourced to subcontractors with the necessary resources. In order to focus on the role of outsourcing, we assume that all the tasks are outsourced. For ease of exposition, we further assume that each task is undertaken by a different subcontractor ("he"), who is selected by the prime contractor for that task as a result of a tendering process.

The prime contractor signs a two-part contract with each subcontractor, which specifies:

1. The time required to complete the task, as well as the resources committed to processing the task, submitted as a tender by each subcontractor.
2. The payment for the task, as well as the option to complete the task earlier, including:

- a) a bonus, payable by the prime contractor to the subcontractor, for completing the task before its normal completion time,
- b) a compensation, payable by the prime contractor to the subcontractor, for crashing the task below its normal duration, and
- c) the allowable ranges of crashing and task completion time.

Part 1 is established according to the subcontractor's tender, which forms a contract once the subcontractor is chosen. Part 2 is specified by the prime contractor. Item 2a specifies an incentive for the subcontractor to complete the task earlier. This may be achieved either by crashing the task, or by starting the task earlier where the predecessors of the task are completed earlier. Item 2b defines the compensation for the crashing cost of the subcontractor, if crashing is performed. Item 2c avoids over-crashing or unnecessary early completion by the subcontractor to exploit the incentives provided by the prime contractor.

We assume that there is a completion cost at each task for the prime contractor, which is a nondecreasing function of the completion time of the task. This cost represents the various bonuses and penalties associated with the task, according to the contract between the prime contractor and the project owner. It includes, for example, the milestone payments discussed in Section 1. A task may be completed earlier so as to reduce its completion cost associated with time; however, this typically requires additional compensation or bonus to motivate the relevant subcontractor. The prime contractor's objective is to *maximize her total profit from com-*

pleting the project. Given the contract that she has signed with the project owner, this is equivalent to minimizing the total cost of completing the project. Henceforth, we consider the cost minimization problem.

The prime contractor aims to achieve her objective by establishing suitable outsourcing contracts with the subcontractors. The development of these contracts requires two types of information to be available to the prime contractor: (i) the costs and resource consumption rates for crashing the tasks, and (ii) the resources that are available to the subcontractors. Crashing usually requires resources such as additional labor and/or equipment to ac-

celerate the processing of a task, and we assume that the resource requirements and costs for these are public information in the marketplace, which is known to all the players. The available resources of the subcontractors are part of the subcontractors' commitment to the project specified in their tenders, which is thus also information that is known to the prime contractor.

Let $\mathcal{N} = \{1, \dots, n\}$ denote the set of subcontractors, or equivalently the set of tasks. Without loss of generality, we assume that task 1 is a dummy start task without predecessors, and n is a dummy end task without successors. We define the following notation.

Parameters :

d_i :	Normal duration of task i
t_i^0 :	Normal completion time of task i , as specified in the contract between the prime contractor and subcontractor i
γ_i :	Normal payment to subcontractor i if task i is completed at time t_i^0
Θ_i :	Unit crashing cost of task i
$\mathcal{R}$:	Set of all types of resources, where $m = \mathcal{R} $
r_{ij} :	Units of resource j consumed by crashing task i by one time unit
$\bar{r}_{ij}$:	Upper bound on the resource j available to subcontractor i
$\phi_i t_i$:	Cost incurred by the prime contractor if task i is completed at time t_i , according to the contract between the prime contractor and the project owner
$\mathcal{P}^i$:	Set of tasks that are immediate predecessors of task i
$\mathcal{A}^i$:	Set of tasks that are immediate successors of task i
D :	Deadline of the project
$v(\mathcal{S})$:	Maximum total profit if all subcontractors in coalition $\mathcal{S}$ act optimally
$c(\mathcal{S})$:	Minimum total cost if all subcontractors in coalition $\mathcal{S}$ act optimally
λ_i :	Profit reallocated by the grand coalition $\mathcal{N}$ to subcontractor i .

Decision variables for the subcontractors :

x_i :	Amount of time by which to crash task i
t_i :	Expedited completion time of task i , where $t_i \leq t_i^0$.

Decision variables for the prime contractor :

- ϑ_i : Cost to be reimbursed to subcontractor i if task i is crashed by one unit of time
- φ_i : Bonus to be paid to subcontractor i for completing task i one unit of time earlier
- $\bar{x}_i$: Maximum allowable amount of time by which task i can be crashed
- $\underline{t}_i$: Earliest allowable completion time of task i .

2.2 Subcontractors' Problem

Suppose that the prime contractor has offered contracts to the subcontractors with parameters ϑ_i , φ_i , $\bar{x}_i$, and $\underline{t}_i$ specified, for all i . Also specified is a crashing cost, θ_i , to be borne by subcontractor i . This is the residual amount of the crashing cost Θ_i after deducting the prime contractor's share ϑ_i , i.e., $\theta_i = \Theta_i - \vartheta_i$. Then, a coalition S of subcontractors, where $|S| \geq 1$, solves the following optimization problem:

$$v(S) = \max_{\{x_i, t_i\}} \sum_{i \in S} \{\varphi_i(t_i^0 - t_i) - \theta_i x_i + \gamma_i\} \quad (1)$$

s.t.

$$\begin{cases} t_i - d_i + x_i \geq t_j, & \forall i \in S, j \in \mathcal{P}^i \cap S \\ t_i - d_i + x_i \geq t_j^0, & \forall i \in S, \text{ with } j \in \mathcal{P}^i \setminus S \\ t_i \leq t_i^0, & \forall i \in S, \text{ with } j \in \mathcal{A}^i \setminus S \end{cases} \quad (2)$$

$$\begin{cases} t_1 - d_1 + x_1 \leq 0, & \text{if } 1 \in S \\ t_1 - d_1 + x_1 \geq 0, & \text{if } 1 \notin S \\ t_n \leq D, & \text{if } n \in S \end{cases} \quad (3)$$

$$\sum_{i \in S} r_{ij} \cdot x_i \leq \sum_{i \in S} \bar{r}_{ij}, \quad \forall j \in \mathcal{R} \quad (4)$$

$$0 \leq x_i \leq \bar{x}_i, \quad \forall i \in S \quad (5)$$

$$t_i \geq \underline{t}_i, \quad \forall i \in S \quad (6)$$

We denote this problem by SC(S). The objective (1) includes the normal payment, the earlier completion time incentive, and the part of the crashing cost that a subcontractor bears. Constraints (2) describe the precedence relations among the tasks, constraints (3) define

the project start time and completion time restrictions, and constraints (4) model resource usage.

Let $\mathbf{x}_N = (x_1, x_2, \dots, x_n)^\top$, $\mathbf{t}_N = (t_1, t_2, \dots, t_n)^\top$, $\mathbf{t}_N^0 = (t_1^0, t_2^0, \dots, t_n^0)^\top$, and $\mathbf{r}_{Nj} = (r_{1j}, r_{2j}, \dots, r_{nj})^\top$, $\forall j \in \mathcal{R}$. Further, let $\mathbf{R}_N^\top = (\mathbf{r}_{N1}, \mathbf{r}_{N2}, \dots, \mathbf{r}_{Nm})$, $\bar{\mathbf{r}}_N = (\sum_{i \in N} \bar{r}_{i1}, \sum_{i \in N} \bar{r}_{i2}, \dots, \sum_{i \in N} \bar{r}_{im})^\top$, $\bar{\mathbf{x}}_N = (\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n)^\top$, $\underline{\mathbf{t}}_N = (\underline{t}_1, \underline{t}_2, \dots, \underline{t}_n)^\top$, $\varphi_N = (\varphi_1, \varphi_2, \dots, \varphi_n)^\top$, $\theta_N = (\theta_1, \theta_2, \dots, \theta_n)^\top$, and $\gamma_N = (\gamma_1, \gamma_2, \dots, \gamma_n)$. Define $\mathbf{0} = (0, 0, \dots, 0)^\top$, $\mathbf{1} = (1, 1, \dots, 1)^\top$, and $\mathbf{I}$ as an identity matrix of appropriate dimension. Then, SC(S) can be rewritten as:

$$v(S) = \max_{\{\mathbf{x}_S, \mathbf{t}_S\}} \varphi_S^\top (\mathbf{t}_S^0 - \mathbf{t}_S) - \theta_S^\top \mathbf{x}_S + \mathbf{1}^\top \gamma_S \quad (7)$$

s.t.

$$\mathbf{A}_S^1 \mathbf{x}_S + \mathbf{A}_S^2 \mathbf{t}_S \leq \mathbf{b}_S^1 \quad (8)$$

$$\mathbf{B}_S^1 \mathbf{x}_S + \mathbf{B}_S^2 \mathbf{t}_S \leq \mathbf{b}_S^2 \quad (9)$$

$$\mathbf{R}_S \mathbf{x}_S \leq \bar{\mathbf{r}}_S \quad (10)$$

$$\mathbf{I} \mathbf{x}_S \leq \bar{\mathbf{x}}_S \quad (11)$$

$$\mathbf{I} \mathbf{t}_S \geq \underline{\mathbf{t}}_S \quad (12)$$

$$\mathbf{x}_S \geq \mathbf{0}; \quad \mathbf{t}_S \geq \mathbf{0} \quad (13)$$

where $\mathbf{A}_S^i$, $\mathbf{B}_S^i$, and $\mathbf{b}_S^i$, $i = 1, 2$, are appropriately defined matrices and vectors that represent Constraints (2) and (3). An example appears in Appendix A.

Two benefits can be attained by members of a cooperating coalition. First, they can share

resources to achieve a more cost efficient crashing solution. Second, they can coordinate their task completion times, so as to benefit from the early completion bonus provided by the prime contractor.

2.3 Prime Contractor's Problem

The prime contractor incurs the following costs: (a) the completion time cost $\phi_i t_i$ for each task i , (b) the payment γ_i to subcontractor i if task i is completed no later than t_i^0 , (c) the per unit time bonus φ_i paid to subcontractor i if task i is completed earlier, i.e., $t_i < t_i^0$, and (d) her contribution ϑ_i to the cost of crashing task i . If the prime contractor sets $\varphi_i = \vartheta_i = 0$, $\forall i \in \mathcal{N}$, then all the tasks are completed normally without any crashing. However, the corresponding completion costs can be rather high. In order to minimize her total cost, the prime contractor needs to find optimal values of φ_i , ϑ_i , $\bar{x}_i$ and $\underline{t}_i$, and specify them in the outsourcing contract with each subcontractor i . The prime contractor minimizes her total cost using the following steps:

1. She collects information about the crashing and resource parameters, i.e. Θ_i , r_{ij} , and $\bar{r}_{ij}$, respectively. She estimates the crashing parameters Θ_i and r_{ij} according to general information available in the marketplace, and receives information about resource availability from the subcontractors as part of their tenders.
2. In order to determine $\bar{x}_i$ and $\underline{t}_i$, the prime contractor solves the following problem, denoted by PC($\mathcal{N}$), to find the optimal crashing solution:

$$\min_{\{\underline{t}_N, \mathbf{x}_N\}} \phi_N^\top \underline{t}_N + \Theta_N^\top \mathbf{x}_N \quad (14)$$

s.t.

$$(8) - (10), (13)$$

where $\phi_N = (\phi_1, \phi_2, \dots, \phi_n)^\top$ and $\Theta_N = (\Theta_1, \Theta_2, \dots, \Theta_n)^\top$. If the optimal solution for problem (14) is denoted by $(\underline{t}_N^*, \mathbf{x}_N^*)$, she sets $\underline{t}_N = \underline{t}_N^*$ and $\bar{x}_N = \mathbf{x}_N^*$. The objective function $\phi_N^\top \underline{t}_N + \Theta_N^\top \mathbf{x}_N$ represents her total cost associated with time for completion, including the normal payment and also compensations for earlier completion or crashing for each task.

3. Let $\vartheta_N = (\vartheta_1, \vartheta_2, \dots, \vartheta_n)^\top$. For given $(\varphi_N, \vartheta_N, \underline{t}_N, \bar{x}_N)$, if all the subcontractors cooperate, then they solve SC($\mathcal{N}$). Suppose that the optimal solution for SC($\mathcal{N}$) is $(\mathbf{t}_N, \mathbf{x}_N)$. Then, the prime contractor solves the following problem to determine the optimal values of $\{\varphi_N, \vartheta_N\}$ that minimize her total cost:

$$\min_{\{\varphi_N, \vartheta_N\}} \{\phi_N^\top \underline{t}_N + \varphi_N^\top (\mathbf{t}_N^0 - \mathbf{t}_N) + \vartheta_N^\top \mathbf{x}_N + \mathbf{1}^\top \gamma_N\} \quad (15)$$

Note that problem (15) above is not a linear optimization problem, since $\mathbf{x}_N$ and $\mathbf{t}_N$ are implicit functions of $\{\varphi_N, \vartheta_N\}$. We discuss how to solve this problem in Section 4.

3. Subcontractors' Cooperative Game

In this section, we consider the strategy of cooperation among the subcontractors. For any coalition $\mathcal{S} \subseteq \mathcal{N}$, its characteristic function $v(\mathcal{S})$ is defined as the maximum total objective value of problem SC($\mathcal{S}$). A cooperative game $(\mathcal{N}, v)$ is thus defined.

3.1 Nonconvexity

A cooperative game is convex if and only if, for any coalitions $\mathcal{S}, \mathcal{T} \subseteq \mathcal{N}$,

$$v(\mathcal{S} \cup \mathcal{T}) + v(\mathcal{S} \cap \mathcal{T}) \geq v(\mathcal{S}) + v(\mathcal{T}) \quad (16)$$

The following example shows that (16) does not hold in our problem, and hence the project game is not convex.

Example 1 Consider the project shown in Figure 1, where $\mathcal{N} = \{1, 2, 3, 4, 5\}$, and $t_i = 2, i \in \mathcal{N}$. There is no crashing cost for any task. The maximum crashing time allowed for each task is $\bar{x}_i = 1$. There is only one type of resource, and $r_{i1} = 1, i \in \mathcal{N}$. The maximum resources which the subcontractors can use are $\bar{r}_{11} = \bar{r}_{51} = 2, \bar{r}_{21} = \bar{r}_{31} = 0.5, \bar{r}_{41} = 1.5$, respectively. Finally, $\varphi_i = 1$ and $\gamma_i = \gamma$, for all $i \in \mathcal{N}$. The normal completion times of the tasks are $\mathbf{t}_N^0 = (2, 4, 4, 4, 6)$, respectively.

Consider coalitions $\mathcal{S} = \{2, 4\}$ and $\mathcal{T} = \{3, 4\}$. Since both coalitions have enough resources for optimal crashing, we have

$$v(\mathcal{S}) = v(\mathcal{T}) = 2[\gamma + 1 \cdot (4 - 3)] = 2\gamma + 2$$

Also, task 4 has enough resources to crash up to its limit by itself, thus

$$v(\mathcal{S} \cap \mathcal{T}) = v(\{4\}) = \gamma + 1 \cdot (4 - 3) = \gamma + 1$$

However, since coalition $\mathcal{S} \cup \mathcal{T}$ has only 2.5 units of resource, its value function is

$$\begin{aligned} v(\mathcal{S} \cup \mathcal{T}) &= v(\{2, 3, 4\}) \\ &= 2[\gamma + 1 \cdot (4 - 3)] + [\gamma + 1 \cdot (4 - 3.5)] \\ &= 3\gamma + 2.5 \end{aligned}$$

Thus,

$$\begin{aligned} v(\mathcal{S} \cup \mathcal{T}) + v(\mathcal{S} \cap \mathcal{T}) &= 4\gamma + 3.5 \\ &< 4\gamma + 4 = v(\mathcal{S}) + v(\mathcal{T}) \end{aligned}$$

which contradicts (16). Hence, the project game is not convex.

Since the game is not convex, some widely used benefit allocation schemes, including the Shapley value (Shapley 1953), may not belong to the core of the game (Peleg and Sudhölter 2003). Without using a core allocation scheme, cooperation may be unstable. However, in Section 3.2, we show that the project game has a nonempty core, and furthermore a profit allocation scheme that belongs to the core can be found by solving a linear program.

3.2 Profit Sharing Scheme

Since problem SC(S) is a linear program, it is natural to consider whether the results of Owen (1975) can be generalized. We note that our subcontractors' cooperative game is more general than Owen's game. The linear program for any coalition $\mathcal{S}$ in Owen's problem takes the form: $\{v(\mathcal{S}) = \max \mathbf{c}^\top \mathbf{x}, \text{ s.t. } \mathbf{B}\mathbf{x} \leq \mathbf{b}(\mathcal{S})\}$, where $\mathbf{b}(\mathcal{S})$ represents the total resources that the coalition $\mathcal{S}$ possesses. Thus, the decision variables $\mathbf{x}$ as well as the resource consumption matrix $\mathbf{B}$ remain the same for different coalitions in Owen's problem. The subcontractors' cooperative game we consider here is different, since each coalition $\mathcal{S}$ has its own decision variables for crashing and its own resource consumption matrix. We can, nevertheless, still apply Owen's idea of using the dual problem.

Let the dual variables corresponding to constraints (8) be denoted by $\rho_N = (\dots, \rho_{ij}, \dots)^\top, i \in \mathcal{N}, j \in \mathcal{P}^i$. Also, let the dual variables corresponding to constraints (9) and (10) be denoted by $\tau_N = (\tau_1, \tau_2, \tau_3)$ and $\omega = (\omega_1, \dots, \omega_m)$, respectively. Finally, let

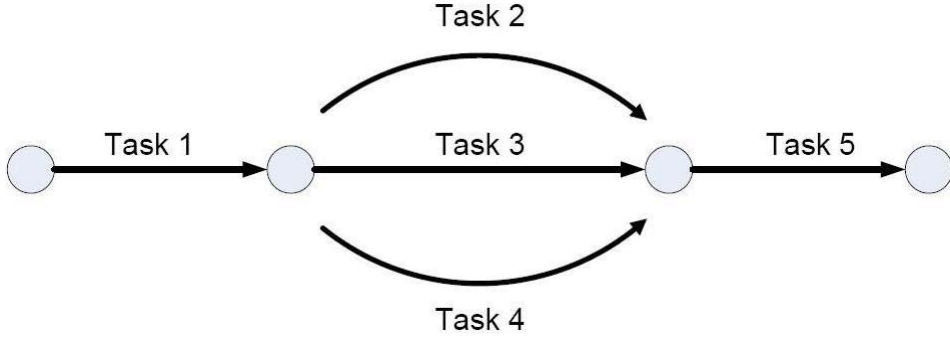


Figure 1 Example Project

the dual variables corresponding to constraints (11) and (12) be denoted by $\mu_N^\top$ and $\zeta_N^\top$, respectively. For an example of the dual problem, see Appendix B. Denote $\mathbf{z}_S^\top = (\rho_S^\top, \tau_S^\top, \omega^\top, \mu_S^\top, \zeta_S^\top)$. With these definitions, the dual of problem

$SC(S)$, denoted by $SD(S)$, can be written as:

$$\min_{\mathbf{z}_S} \quad (\mathbf{b}_S^1)^\top \rho_S + (\mathbf{b}_S^2)^\top \tau_S + \bar{\mathbf{r}}_S^\top \omega + \bar{\mathbf{x}}_S^\top \mu_S - \bar{\mathbf{t}}_S^\top \zeta_S + \varphi_S^\top \mathbf{t}_S^0 + \mathbf{1}^\top \gamma_S \quad (17)$$

s.t.

$$\begin{pmatrix} (\mathbf{A}_S^1)^\top & (\mathbf{B}_S^1)^\top & \mathbf{R}_S^\top & \mathbf{I} & \mathbf{0} \\ (\mathbf{A}_S^2)^\top & (\mathbf{B}_S^2)^\top & \mathbf{0} & \mathbf{0} & -\mathbf{I} \end{pmatrix} \begin{pmatrix} \rho_S \\ \tau_S \\ \omega \\ \mu_S \\ \zeta_S \end{pmatrix} \geq \begin{pmatrix} -\theta_S \\ -\varphi_S \end{pmatrix} \quad (18)$$

$$\mathbf{z}_S \geq \mathbf{0} \quad (19)$$

Denote the coefficient vector in the objective function of $SD(S)$ by $\mathbf{b}_S$. Then, we have

$$\mathbf{b}_S^\top = ((\mathbf{b}_S^1)^\top, (\mathbf{b}_S^2)^\top, \bar{\mathbf{r}}_S^\top, \bar{\mathbf{x}}_S^\top, -\bar{\mathbf{t}}_S^\top). \quad (20)$$

Further, we use a similar definition of $\mathbf{b}_{\{i\}}$ for any coalition $S = \{i\}$ with a single member i .

Now, let $\mathbf{z}_S^*$ and $\mathbf{z}_N^*$ be the optimal solutions for $SD(S)$ and $SD(N)$, respectively. In order to identify an allocation scheme that is in the core, we first establish the following two results.

Lemma 1

$$\mathbf{b}_S^\top \mathbf{z}_S^* \leq \mathbf{b}_S^\top \mathbf{z}_N^*$$

Proof. We prove the lemma by showing that the optimal solution $\mathbf{z}_N^*$ of $SD(N)$ is a feasible

solution of $SD(S)$. First, consider the matrix $((\mathbf{A}_N^1)^\top | (\mathbf{B}_N^1)^\top)$ in the dual problem $SD(N)$. Row i corresponds to a decision variable x_i in problem $SC(N)$. Thus, if task i does not belong to coalition S , then $((\mathbf{A}_S^1)^\top | (\mathbf{B}_S^1)^\top)$ does not contain row i of $((\mathbf{A}_N^1)^\top | (\mathbf{B}_N^1)^\top)$. Similarly, row i of the matrix $((\mathbf{A}_N^2)^\top | (\mathbf{B}_N^2)^\top)$ corresponds to the primal variable t_i . If task i does not belong to the coalition S , then $((\mathbf{A}_S^2)^\top | (\mathbf{B}_S^2)^\top)$ does not contain row i of $((\mathbf{A}_N^2)^\top | (\mathbf{B}_N^2)^\top)$. A similar relationship exists between $\mathbf{r}_N$ and $\mathbf{r}_S$.

Second, any nonzero column of the matrices $((\mathbf{A}_S^k)^\top | (\mathbf{B}_S^k)^\top)$, $k = 1, 2$, must correspond to a constraint in the primal problem $SC(N)$

that includes at least one of the primal variables x_i and t_i of task $i \in \mathcal{S}$. Moreover, those columns in $\left((\mathbf{A}_N^k)^\top | (\mathbf{B}_N^k)^\top\right)$ that correspond to constraints involving no primal variables in $\mathcal{S}$ appear in $\left((\mathbf{A}_S^k)^\top | (\mathbf{B}_S^k)^\top\right)$ as zero columns. A similar observation can be made for the identity matrices $\mathbf{I}_N$ and $\mathbf{I}_S$.

Let $\mathbf{E}_N$ and $\mathbf{E}_S$ denote the LHS matrices of constraints (18) of the dual problems $\text{SD}(\mathcal{N})$ and $\text{SD}(\mathcal{S})$, respectively. It follows from the above observations that, by

(i) deleting those rows i of $\mathbf{E}_N$ that correspond to the tasks $i \notin \mathcal{S}$, and

(ii) changing those columns j of $\mathbf{E}_N$ that correspond to the constraints which do not involve primal variables in the coalition $\mathcal{S}$ into zero columns, we obtain $\mathbf{E}_S$.

Finally, after deleting all the i th components of $\theta_N^\top$ and $\varphi_N^\top$, for $i \notin \mathcal{S}$, we obtain $\theta_S^\top$ and $\varphi_S^\top$.

It follows that the constraints of $\text{SD}(\mathcal{S})$ are a subset of the constraints of $\text{SD}(\mathcal{N})$. Therefore, any solution $\mathbf{z}_N$ satisfying the constraints of $\text{SD}(\mathcal{N})$ also satisfies the constraints of $\text{SD}(\mathcal{S})$. ■

An example of the construction used in Lemma 1 appears in Appendix B.

Lemma 2 For any coalition $\mathcal{S}$,

$$\mathbf{b}_S = \sum_{i \in \mathcal{S}} \mathbf{b}_{\{i\}} \quad (21)$$

Proof. From (20) and the definition of $\mathbf{b}_{\{i\}}$, we need only show that $\mathbf{b}_S^1 = \sum_{i \in \mathcal{S}} \mathbf{b}_{\{i\}}^1$ and $\mathbf{b}_S^2 = \sum_{i \in \mathcal{S}} \mathbf{b}_{\{i\}}^2$, since the equality of the other elements between $\mathbf{b}_S$ and $\sum_{i \in \mathcal{S}} \mathbf{b}_{\{i\}}$ is clear.

Consider the k th element of $\mathbf{b}_S^1$ that corresponds to the precedence relation between

tasks i and j , where $i \in \mathcal{S}$. Then, we have the following results:

- (A) If $j \in \mathcal{P}^i \cap \mathcal{S}$, then the k th element of $\mathbf{b}_S^1$ equals $-d_i$. Moreover, the k th element of $\sum_{h \in \mathcal{S}} \mathbf{b}_{\{h\}}^1$ equals the sum of the k th elements of $\mathbf{b}_{\{i\}}^1$ and $\mathbf{b}_{\{j\}}^1 = (-d_i - t_j^0) + t_j^0 = -d_i$.
- (B) If $j \in \mathcal{P}^i \setminus \mathcal{S}$, then the k th element of $\mathbf{b}_S^1$ equals $-d_i - t_j^0$. Moreover, the k th element of $\sum_{k \in \mathcal{S}} \mathbf{b}_{\{k\}}^1$ equals the k th element of $\mathbf{b}_{\{i\}}^1 = -d_i - t_j^0$.
- (C) If $j \in \mathcal{A}^i \cap \mathcal{S}$, then the k th element of $\mathbf{b}_S^1$ equals the sum of the k th elements of $\mathbf{b}_{\{i\}}^1$ and $\mathbf{b}_{\{j\}}^1 = t_i^0 + (-d_j - t_i^0) = -d_j$.
- (D) If $j \in \mathcal{A}^i \setminus \mathcal{S}$, then the k th element of $\mathbf{b}_S^1$ equals t_i^0 . Moreover, the k th element of $\sum_{k \in \mathcal{S}} \mathbf{b}_{\{k\}}^1$ equals the k th element of $\mathbf{b}_{\{i\}}^1 = t_i^0$.

It follows from cases (A)–(D) above that $\mathbf{b}_S^1 = \sum_{i \in \mathcal{S}} \mathbf{b}_{\{i\}}^1$. Since $\mathbf{b}_N^2 = (d_1, -d_1, D)^\top$, we have $\mathbf{b}_S^2 = \sum_{i \in \mathcal{S}} \mathbf{b}_{\{i\}}^2$, according to whether task 1 or task n belongs to the coalition $\mathcal{S}$. ■

We note that $\mathbf{b}_S^\top \mathbf{z}_S^*$ is the extra benefit generated from cooperation of members in the coalition $\mathcal{S}$. Lemma 2 implies that the total extra profit of the coalition is the sum of the extra profits of its members. We now provide the main result of this section.

Theorem 1 Let $\lambda = (\lambda_1, \dots, \lambda_n)$, where

$$\lambda_i = \mathbf{b}_{\{i\}}^\top \mathbf{z}_N^* + \varphi_i t_i^0 + \gamma_i \quad (22)$$

Then, λ constitutes a core allocation of the game $(\mathcal{N}, v)$.

Proof. Let $\mathbf{t}_S^*, \mathbf{x}_S^*$ be an optimal solution of $\text{SC}(\mathcal{S})$, and let $v^*(\mathcal{S})$ be the corresponding optimal objective value. For any coalition $\mathcal{S} \subseteq \mathcal{N}$,

we have

$$\begin{aligned}
\sum_{i \in S} \lambda_i &= \sum_{i \in S} [\mathbf{b}_{\{i\}}^\top \mathbf{z}_N^* + \varphi_i t_i^0 + \gamma_i] \\
&= \mathbf{b}_S^\top \mathbf{z}_N^* + \varphi_S^\top \mathbf{t}_S^0 + \mathbf{1}^\top \gamma_S \\
&\geq \mathbf{b}_S^\top \mathbf{z}_S^* + \varphi_S^\top \mathbf{t}_S^0 + \mathbf{1}^\top \gamma_S \\
&= \varphi_S^\top (\mathbf{t}_S^0 - \mathbf{t}_S^*) - \theta_S^\top \mathbf{x}_S^* + \mathbf{1}^\top \gamma_S \\
&= v^*(S)
\end{aligned}$$

where the second equality follows from Lemma 2, the inequality follows from Lemma 1, and the third equality follows from strong duality. It follows that $\lambda = (\lambda_1, \dots, \lambda_n)$ lies in the core of the game (N, v) . ■

Theorem 1 implies that the game is balanced, hence for any instance there exists an incentive scheme that is sufficient for the subcontractors to cooperate within the grand coalition N . Moreover, the theorem specifies a complete profit distribution scheme which allocates λ_i , as now defined, to subcontractor i .

Remark 1 Equation (22) is equivalent to

$$\lambda_i = \gamma_i + \varphi_i t_i^0 + \sum_{j=1}^m \bar{r}_{ij} \omega_j^* - \underline{t}_i \zeta_i^* + \bar{x}_i \mu_i^* + \begin{cases} t_1^0 \sum_{k \in \mathcal{A}^1} \rho_{k1}^* + d_1(\tau_1^* - \tau_2^*), & i = 1 \\ t_i^0 \sum_{k \in \mathcal{A}^i} \rho_{ki}^* - (d_i + t_j^0) \sum_{j \in \mathcal{P}^i} \rho_{ij}^*, & i = 2, \dots, n-1 \\ D\tau_3^* - (d_n + t_j^0) \sum_{j \in \mathcal{P}^n} \rho_{nj}^*, & i = n \end{cases}$$

Appendix C provides an example of the payoff distribution, λ .

3.3 Task Lengthening

We now consider a generalization of the problem in Section 2.2, where the tasks can be lengthened to release extra resources for use by other subcontractors to crash their tasks. This possibility can increase the value of a co-operating coalition of subcontractors.

Define $\mathbf{y}_N = (y_1, y_2, \dots, y_n)^\top$, $\bar{\mathbf{y}}_N = (\bar{y}_1, \bar{y}_2, \dots, \bar{y}_n)^\top$, and $\bar{\mathbf{t}}_N = (\bar{t}_1, \bar{t}_2, \dots, \bar{t}_n)^\top$ where y_i denotes the number of time units by which task i is lengthened, $\bar{y}_i$ denotes the maximum amount by which task i can be lengthened and $\bar{t}_i$ denotes the latest completion time of task i . Problem $SC(N)$ can be rewritten as

$$v(N) = \max_{\{\mathbf{x}_N, \mathbf{y}_N, \mathbf{t}_N\}} \varphi_N^\top (\mathbf{t}_N^0 - \mathbf{t}_N) - \theta_N^\top \mathbf{x}_N + \mathbf{1}^\top \gamma_N \quad (23)$$

s.t.

$$\mathbf{A}_N^1(\mathbf{x}_N - \mathbf{y}_N) + \mathbf{A}_N^2 \mathbf{t}_N \leq \mathbf{b}_N^1$$

$$\mathbf{B}_N^1(\mathbf{x}_N - \mathbf{y}_N) + \mathbf{B}_N^2 \mathbf{t}_N \leq \mathbf{b}_N^2$$

$$\mathbf{R}_N(\mathbf{x}_N - \mathbf{y}_N) \leq \bar{\mathbf{r}}_N$$

$$\mathbf{I} \mathbf{x}_N \leq \bar{\mathbf{x}}_N; \mathbf{I} \mathbf{y}_N \leq \bar{\mathbf{y}}_N$$

$$\mathbf{I} \mathbf{t}_N \leq \bar{\mathbf{t}}_N; \mathbf{I} \mathbf{t}_N \geq \underline{\mathbf{t}}_N$$

$$\mathbf{x}_N \geq \mathbf{0}; \mathbf{y}_N \geq \mathbf{0}; \mathbf{t}_N \geq \mathbf{0}$$

The main result of this section now follows.

The proof follows from that of Theorem 1.

Theorem 2 Let $\lambda = (\lambda_1, \dots, \lambda_n)$, where

$$\lambda_i = \hat{\mathbf{b}}_{\{i\}}^\top \hat{\mathbf{z}}_N^* + \varphi_i t_i^0 + \gamma_i \quad (24)$$

$\hat{\mathbf{b}}_{\{i\}}$ is the vector $\mathbf{b}_{\{i\}}$ augmented with $(0, \dots, \bar{\mathbf{y}}^\top, \dots, 0)^\top$, and $\hat{\mathbf{z}}_N^*$ is the optimal dual solution of the problem $SC(N)$. Then, λ constitutes a core allocation of the game (N, v) with task lengthening.

Appendix D provides an example of the project game with task lengthening.

4. Contract Design

In this section, we investigate the optimal subcontract design problem of the prime contractor. We model this problem as a Stackelberg game, where the prime contractor is the leader

of the game who specifies contracts by anticipating the reactions of the subcontractors, and the subcontractors are followers who attempt to maximize their profits, given the contracts.

Recall that t_i^0 is the normal completion time of task i and $\phi_i t_i^0$ is the corresponding completion time cost for the prime contractor. If task i is completed at an earlier time $t_i < t_i^0$, then the saving in the completion time cost is $\phi_i(t_i^0 - t_i)$. The prime contractor may share this saving with the subcontractor by offering him a bonus $\alpha_i \phi_i(t_i^0 - t_i)$, hence $\varphi_i = \alpha_i \phi_i$. Consequently, the saving that the prime contractor retains is $(1 - \alpha_i) \phi_i(t_i^0 - t_i)$. The prime contractor may also share a portion of the subcontractor's crashing cost if his task is crashed. Suppose that the per unit crashing cost for task i is Θ_i , and the prime contractor shares $\beta_i \Theta_i$. Then, $\vartheta_i = \beta_i \Theta_i$ and $\theta_i = (1 - \beta_i) \Theta_i$. The contract design problem for the prime contractor is to determine the contract parameters $\{\alpha_i\}, \{\beta_i\}, \{\bar{x}_i\}, \{\underline{t}_i\}$ that maximize her overall profit.

4.1 Uniform Contracts

We first study the special case of uniform contracts, where $\alpha_i = \alpha$ and $\beta_i = \beta$, for all i . In this case, the prime contractor offers the same contract to all the subcontractors. Using uniform contracts is administratively simple for the prime contractor, and avoids any legal concerns about discriminatory pricing. We further assume that $\alpha = 1 - \beta$. This implies that the proportion of earlier completion time bonus and the proportion of crashing cost shared by the prime contractor are the same. Equivalently, the proportion of cost savings received by the subcontractors is the same, irrespective

of whether they contribute by early completion or by crashing. We also assume a lower bound $\underline{\alpha}$ for α , representing the minimum share of profit that the prime contractor must give to a subcontractor in exchange for his cooperation.

The prime contractor first solves problem $PC(N)$, as described in Step 2 in Section 2.3, to find the minimum cost solution $\mathbf{x}_N^*$ and $\mathbf{t}_N^*$, and then sets the allowable ranges for the crashing amounts and completion times as $\bar{\mathbf{x}}_N^* = \mathbf{x}_N^*$ and $\underline{\mathbf{t}}_N^* = \mathbf{t}_N^*$. We show below that this design of $\bar{\mathbf{x}}_N^*$ and $\underline{\mathbf{t}}_N^*$ allows optimization of the prime contractor's contract. Intuitively, the prime contractor needs to impose upper bounds on the crashing amounts and lower bounds on the completion times in the contracts with the subcontractors, in order to prevent the latter taking advantage of the incentives she offers.

Given α and $\bar{\mathbf{x}}_N^*$ and $\underline{\mathbf{t}}_N^*$, the problem for the grand coalition $SC(N)$ of the subcontractors is:

$$v(N) = \max_{\{\mathbf{x}_N, \mathbf{t}_N\}} \alpha \phi_N^T (\mathbf{t}_N^0 - \mathbf{t}_N) + \mathbf{1}^T \gamma_N - \alpha \Theta_N^T \mathbf{x}_N \quad (25)$$

s.t.

$$(8) - (10), (13)$$

$$\mathbf{I} \mathbf{x}_N \leq \bar{\mathbf{x}}_N^* \quad (26)$$

$$\mathbf{I} \mathbf{t}_N \geq \underline{\mathbf{t}}_N^* \quad (27)$$

We denote the problem above by $\overline{SC}(N)$ and its optimal solution by $\{\mathbf{x}'_N, \mathbf{t}'_N\}$.

Lemma 3 *The optimal solution $\{\mathbf{x}'_N, \mathbf{t}'_N\}$ of problem $\overline{SC}(N)$ is also an optimal solution of problem $PC(N)$.*

Proof. Since $\mathbf{1}^T \gamma_N$ and $\phi_N^T \mathbf{t}_N^0$ are constants, the objective function (25) is equivalent to minimizing $\alpha(\phi_N^T \mathbf{t}_N + \Theta_N^T \mathbf{x}_N)$, i.e., the objective function (14). Therefore, an optimal solution

$\{\mathbf{x}'_N, \mathbf{t}'_N\}$ of problem $\overline{\text{SC}}(N)$ is an optimal solution of problem $\text{PC}(N)$, subject to the two additional constraints (26) and (27). However, constraints (26) and (27) are not restrictive for problem $\text{PC}(N)$. Hence, $\{\mathbf{x}'_N, \mathbf{t}'_N\}$ is also optimal for problem $\text{PC}(N)$. ■

From the proof above, we observe that

$$\begin{aligned} & \min_{\alpha \geq \underline{\alpha}} \quad \{\alpha \phi_N^\top \mathbf{t}_N^0 + \mathbf{1}^\top \gamma_N + (1 - \alpha)[\Theta_N^\top \mathbf{x}_N^* + \phi_N^\top \mathbf{t}_N^*]\} \\ = & \min_{\alpha \geq \underline{\alpha}} \quad \{\alpha[\phi_N^\top (\mathbf{t}_N^0 - \mathbf{t}_N^*) - \Theta_N^\top \mathbf{x}_N^*] + \mathbf{1}^\top \gamma_N + [\Theta_N^\top \mathbf{x}_N^* + \phi_N^\top \mathbf{t}_N^*]\} \end{aligned}$$

Note that $[\phi_N^\top (\mathbf{t}_N^0 - \mathbf{t}_N^*) - \Theta_N^\top \mathbf{x}_N^*] \geq 0$ because any crashing decision $\mathbf{x}_N^*$ will not increase cost, else the optimal crashing amount is 0. Thus, the optimal α^* must equal $\underline{\alpha}$. We now state our main result for uniform contracts.

Theorem 3 *With uniform contracts, the prime contractor's profit is maximized by setting $\bar{\mathbf{x}}_N = \mathbf{x}_N^*$, $\mathbf{t}_N = \mathbf{t}_N^*$, and $\alpha^* = \underline{\alpha}$, where $\{\mathbf{x}_N^*, \mathbf{t}_N^*\}$ is the optimal solution of problem $\text{PC}(N)$.*

Theorem 3 implies that the prime contractor always receives $1 - \underline{\alpha}$ share of the total profit. If the prime contractor has dominating market power, then she can specify a low value of $\underline{\alpha}$.

4.2 Nonuniform Contracts

Although a uniform contracts scheme is transparently fair to all the subcontractors, it is intuitive that the prime contractor may gain by providing customized incentives to the subcontractors with different α_i and β_i values, especially where a lot of crashing is needed. In this subsection, we assume that there are no legal or administrative restrictions on this.

Task i can be completed earlier in two ways:

(i) task i is crashed; and (ii) its predecessors

$\{\mathbf{x}'_N, \mathbf{t}'_N\}$ is independent of $\alpha > 0$. That is, the optimal solution of the grand coalition of the subcontractors remains the same for any positive incentive $\alpha > 0$ provided by the prime contractor. This, together with Lemma 3, enables us to rewrite the problem of the prime contractor as follows:

are crashed and completed earlier, and so it can be started earlier. In both cases, subcontractor i receives an early completion bonus of $\alpha_i \phi_i$ per unit time. In case (i), the subcontractor's crashing cost is $(1 - \beta_i)\Theta_i$, if the prime contractor compensates an amount $\beta_i \Theta_i$. Therefore, the problem for the prime contractor is to set (α_N, β_N) appropriately such that the subcontractors are motivated to crash their tasks while the prime contractor maximizes her profit. We propose Algorithm AE below to solve this problem. To illustrate our main ideas, we first discuss an example in which five tasks are configured as in Figure 2. For convenience, we assume that the crashing times must take integer values in this example.

Suppose that the normal durations of the five tasks are $d_1 = 3$, $d_2 = 4$, $d_3 = 8$, $d_4 = 2$ and $d_5 = 3$, respectively. If no crashing is conducted on any task, then the completion times of the tasks are $t_1^0 = 3$, $t_2^0 = 7$, $t_3^0 = 11$, $t_4^0 = 9$, and $t_5^0 = 14$. Furthermore, suppose that, after solving the prime contractor's problem $\text{PC}(N)$, the optimal crashing amounts of the tasks are: $x_1^* = 0$, $x_2^* = 1$, $x_3^* = 3$, $x_4^* = 0$, and $x_5^* = 1$. The total amount of crashing is $\sum x_i^* = 5$. We de-

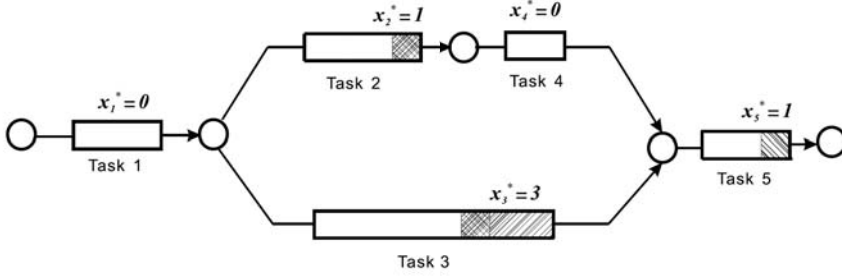


Figure 2 Crashing Operations in an Example Project

fine the crashing operations to be grouped into three "basic crashing subsets" (BCS), as follows (see the shaded areas in Figure 2):

$$C(1) = \{\text{Task 5, crashed by 1 time unit}\}$$

$$C(2) = \{\text{Task 3, crashed by 2 time units}\}$$

$$C(3) = \{\text{Tasks 2 and 3, crashed by 1 time unit simultaneously}\}$$

In general, a BCS is defined as a subset of tasks, all of which must be crashed simultaneously by the same amount, in order to achieve the necessary crashing. If the total allowable crashing time is limited by an upper bound, denoted by X , then the crashing operations over these BCSs should be implemented accordingly as the total allowable crashing time changes. For example, suppose that crashing Task 5 is more beneficial than crashing Task 3, where we define "benefit" as the total saving generated by the crashing minus the total cost to perform the crashing. Further, assume that crashing Task 3 alone is more beneficial than crashing Tasks 2 and 3 simultaneously. Then, if $X = 1$, Task 5 is crashed, i.e. the crashing operation in $C(1)$ is implemented. If X is increased to 3, then Task 5 in $C(1)$ remains crashed by 1 unit, and Task 3 in $C(2)$ is crashed by 2 units. If X is increased to 5, then the crashing operations in $C(1)$ and $C(2)$ remain, and Tasks 2 and 3 are crashed simultaneously by 1 unit.

An intuitive description of our algorithm is as follows:

- We search values of X over its possible range, in order to identify all the BCSs $C(q)$, $q = 1, 2, \dots$.
- We employ a series of bisection searches to determine $C(q)$, $q = 1, 2, \dots$. Using each bisection search, we identify the smallest value of X at which we find a $C(q)$ in which the crashing time of at least one task is crashed to its upper bound x_k^* , which we describe as "saturated".
- For each $C(q)$, we identify a cost-bonus constraint with respect to the parameters (α_N, β_N) , which ensures that the subcontractors perform the necessary crashing.
- Finally, the optimal parameters (α_N^*, β_N^*) are determined by maximizing the total profit of the prime contractor, subject to all the cost-bonus constraints.

For the example in Figure 2, the procedure to determine the BCSs works as follows:

- Initially, we search X over the interval $[0, X_U] = [0, 5]$, where X_U is the maximum total crashing amount. Using bisection search (for details, see Algorithm AE described below), we examine the points $X = 3, 2, 1$. At $X = 1$, $C(1)$ with Task 5 is identified, where Task 5 is satu-

rated. Then, we set $x_5 = 1$.

- We search X over the interval $[1, 5]$. Using bisection search, we examine the points $X = 3, 4, 5$. At $X = 5$, $C(3)$ with Tasks 2 and 3 is identified, where Task 2 is saturated. Then, we keep $x_5 = 1$, and set $x_2 = x_3 = 1$.
- We search X over the interval $[3, 5]$. Using bisection search, we examine the points $X = 4, 5$. At $X = 5$, $C(2)$ with Task 3 is identified, where Task 3 is saturated. Then, we keep $x_2 = x_5 = 1$, and set $x_3 = 1 + 2 = 3$.

For $C(1)$, we define the cost-bonus constraint $\alpha_5\phi_5 \geq (1 - \beta_5)\Theta_5$, which ensures that Subcontractor 5 crashes Task 5. For $C(3)$, the cost-bonus constraint is $(\alpha_2\phi_2 + \alpha_3\phi_3) + (\alpha_4\phi_4 + \alpha_5\phi_5) \geq (1 - \beta_2)\Theta_2 + (1 - \beta_3)\Theta_3$, where $\alpha_4\phi_4$ and $\alpha_5\phi_5$ are the bonuses due to the earlier completions of Task 4 and 5, respectively, achieved as a result of crashing Tasks 2 and 3. Similarly, for $C(2)$, we define the cost-bonus constraint $\alpha_3\phi_3 + \alpha_5\phi_5 \geq (1 - \beta_3)\Theta_3$. The optimal contract design problem for the prime contractor is to determine the optimal parameters (α_i, β_i) , $i = 1, 2, \dots, 5$, to maximize her profit subject to these cost-bonus constraints.

In general, the crashing time for a task can be a nonnegative real number, instead of an integer as assumed in the example in Figure 2. In such a case, a bisection search is applied over a continuous interval $[0, X_U]$, and terminates once the remaining interval of search has width less than or equal to a prespecified number ε . In practice, ε is determined by the smallest implementable unit of crashing. It is well known that the number of iterations to achieve such an interval is equal to

$(\log X_U - \log \varepsilon) / \log 2$. With ε being a constant, the time complexity of the bisection procedure is $O(\log X_U)$.

Formally, we propose the following algorithm.

Algorithm AE:

- Step 1. Solve the optimal crashing problem $PC(N)$ in Section 2.3 to obtain the optimal solution $\mathbf{x}^* = (x_1^*, \dots, x_n^*)$ and $\mathbf{t}^* = (t_1^*, \dots, t_n^*)$. Let $X_a = 0$ and $X_b = X_U$, where $X_U = \sum_{i=1}^n x_i^*$, and let the lower bounds on crashing times x_i be $x_i^L = 0$, $i = 1, \dots, n$. Set $q = 1$.
- Step 2. Perform a bisection search over the interval $[X_a, X_b]$, to search for the smallest X at which a subset $C(q)$ of tasks are crashed with at least one task k saturating its maximum crashing time x_k^* , as follows:

- Step 2a. Let $X = X_a + \frac{1}{2}(X_b - X_a)$. Solve the following linear program with $\mathcal{S} = N$:

$$\min_{\{x_i, t_i\}} \sum_{i \in \mathcal{S}} [\phi_i t_i + \Theta_i x_i] \quad (28)$$

s.t.

$$\left\{ \begin{array}{l} (8) - (10), (13) \\ \sum_{i \in \mathcal{S}} x_i \leq X \\ x_i^L \leq x_i \leq x_i^*, \forall i \in \mathcal{S}. \end{array} \right. \quad (29)$$

Check whether the optimal solution found is unique or not, and execute one of the following two steps accordingly.

- Step 2a.i. There is a unique optimal solution: Let the solution be $\{x'_i, t'_i\}$

and check if there is any task k such that $x'_k = x_k^*$.

Step 2a.ii. There are multiple optimal solutions: Determine the optimal extreme points, and check if at any optimal extreme point, a task k achieves its maximum crashing time, i.e., $x'_k = x_k^*$.

Step 2b. If there exists no task such that $x'_k = x_k^*$, then set $X_a = X$, and go to Step 2a. Otherwise, if there exists at least one such task k , then set $X_b = X$, and go to Step 3 if $X_b - X_a \leq \varepsilon$, where ε is a pre-specified tolerance value, or go to Step 2a otherwise.

s.t.

$$\begin{cases} \sum_{i \in C(j)} \alpha_i \phi_i + \sum_{i \in A(j)} \alpha_i \phi_i \geq \sum_{i \in C(j)} (1 - \beta_i) \Theta_i, & \forall j = 1, \dots, q \\ 0 \leq \alpha_i \leq 1, 0 \leq \beta_i \leq 1, & \forall i \in \mathcal{N} \end{cases} \quad (31)$$

$$(32)$$

to obtain the optimal α_i^*, β_i^* for all i .

Output: The optimal values $x_i^*, t_i^*, \alpha_i^*, \beta_i^*, \forall i \in \mathcal{N}$.

It is possible that the linear program (28)-(29) has multiple optimal solutions. This possibility complicates the algorithm. For example, suppose there are two optimal crashing solutions, (2,0) and (1,1), and suppose the maximum crashing time of task 1 is 2. Then, in the first case, we fix that value of X and identify a BCS. In the second case, we do not, however, so we must next try a smaller value of X . But at the smaller value of X , we may find that no task achieves its crashing limit. This example shows that, when the linear program has a set of multiple optimal solutions, we need to check whether the set contains *any solution* in which

crashed simultaneously with task k as $C(q)$. Update the lower bounds: $x_i^L = x_i^L + x'_k$ for all tasks $i \in C(q)$. Identify the subset $A(q)$ of succeeding tasks that can be started earlier due to the crashed tasks. Go to Step 4 if the crashing times for all tasks have reached $\{x_i^*\}$. Otherwise, let $X_a = \sum_{i=1}^n x_i^L$, and $X_b = X_U$, and go to Step 2 with $q = q + 1$.

Step 4. Solve the following linear program:

$$\max_{\{\alpha_i, \beta_i\}} \sum_{i \in \mathcal{N}} (1 - \alpha_i) \phi_i (t_i^0 - t_i^*) - \beta_i \Theta_i x_i^* \quad (30)$$

at least one task saturates its maximum crashing time. Step 2a.ii is designed to deal with this issue.

First, note that the feasible set of the linear program (28)-(29) is bounded, due to the upper bounds on x_i and the deadline on the completion time. Thus, the optimal solutions of the problem can be expressed as a convex combination of the optimal extreme points; see, for example, Bazaraa et al. (1990). We let $\mathbf{y}^T = (\mathbf{x}^T, \mathbf{t}^T)$, and denote the optimal extreme points as $\mathbf{y}^j, j = 1, 2, \dots, J$, where J is the total number of optimal extreme points. Then, any optimal solution $\mathbf{y}'$ can be expressed as $\mathbf{y}' = \sum_{j=1}^J \kappa_j \mathbf{y}^j$, where

$$\sum_{j=1}^J \kappa_j = 1, \quad \text{and} \quad \kappa_j \geq 0 \quad \text{for all } j \quad (33)$$

What we need in Step 2a above is to check whether the set of optimal solutions contains any solution in which at least one of the tasks saturates its maximum crashing time. The following lemma shows that, to do so, we need only consider the optimal extreme points.

Lemma 4 *The linear program (28)–(29) has an optimal solution at which a task k achieves its maximum crashing amount x_k^* , if and only if the problem has an optimal extreme point at which task k reaches its maximum amount x_k^* .*

Proof. We only need to prove the necessary condition, since the sufficient condition is trivial. Suppose that the problem has a set of optimal solutions. Then, the k th element x'_k of an optimal solution can be expressed as $x'_k = \sum_{j=1}^J \kappa_j x_k^j$, where x_k^j is the crashing amount of task k at the optimal extreme point j . If $x_k^j \neq x_k^*$, for all j , then it is impossible to have $x'_k = x_k^*$ since x_k^* is the upper bound on the decision variable x_k and the coefficients κ_j must satisfy (33). ■

The lemma indicates that we need only check the optimal extreme points without having to consider other points in the set of optimal solutions. This justifies Step 2a.ii. The simplex method can be used to identify all the optimal extreme points (Zuo et al. 2000). In practice, the number of optimal extreme points is typically small, hence this step can be executed with computational efficiency.

We now prove that the α_N and β_N values found by Algorithm AE represent the optimal contract design for the prime contractor. We

use $LP(2n)$ in the theorem below to denote the time complexity of solving the linear program in Step 2a, including the time requirement for checking multiple optimal extreme points if needed.

Theorem 4 *The parameters (α_N, β_N) derived by Algorithm AE together with $\bar{\mathbf{x}}_N = \mathbf{x}_N^*$ and $\underline{\mathbf{t}}_N = \mathbf{t}_N^*$ constitute the optimal contracts of the prime contractor with the subcontractors, which maximize her total profit. Also, the time complexity of Algorithm AE is $O(n \log(\sum_i \bar{x}_i) LP(2n))$.*

Proof. We first show that, under the contracts specified by the theorem: (a) the subcontractors crash and complete their tasks according to $(\mathbf{x}_N^*, \mathbf{t}_N^*)$, and (b) the prime contractor's profit is maximized.

First, observe that any optimal crashing profile of a project problem comprises a number of BCSs. Moreover, there must exist a smallest total crashing time X , at which the crashing of a BCS is implemented with at least one task k saturating its maximum crashing time x_k^* . The bisection search of Step 2 finds a BCS corresponding to the smallest value of X , and is justified by Lemma 4. Since crashing all the tasks in this BCS with the same amount of crashing time x'_k does not affect the freedom of crashing of any other BCS, the updating operation of Step 3 of the algorithm allows us to identify all the BCSs of the problem.

From Theorem 1, we know that the subcontractors cooperate, and thus crashing of any BCS is performed as long as the total benefit achieved by the coalition is not less than the total cost. This is ensured by the cost-bonus constraints (31). As a result, the coalition performs crashing operations of $C(j)$ for $j = 1, \dots, q$,

and reaches the solution $(\mathbf{x}_N^*, \mathbf{t}_N^*)$.

Next, we consider the benefit of the prime contractor from the optimal solution (α_N, β_N) of the linear program (30)–(32). Note that this solution must satisfy (31) as an equality; otherwise, we can decrease β_i so that the constraint (31) is still satisfied and the objective function (30) is increased. Therefore, the prime contractor's benefit from crashing one time unit of the tasks in subset $C(j)$ can be calculated as:

$$\begin{aligned} & \sum_{i \in C(j)} (1 - \alpha_i) \phi_i + \sum_{i \in A(j)} (1 - \alpha_i) \phi_i \\ & - \sum_{i \in C(j)} \beta_i \Theta_i \\ = & \sum_{i \in C(j)} (1 - \alpha_i) \phi_i + \sum_{i \in A(j)} (1 - \alpha_i) \phi_i \\ & + \sum_{i \in C(j)} (1 - \beta_i) \Theta_i - \sum_{i \in C(j)} \Theta_i \\ = & \sum_{i \in C(j)} \phi_i + \sum_{i \in A(j)} \phi_i - \sum_{i \in C(j)} \Theta_i \end{aligned}$$

where the first two terms are the benefits from completing the tasks one unit of time earlier, and the last term is the crashing cost. The total benefit obtained by the prime contractor is the sum of the benefits achieved by the crashing operations of all BCSs, until the solution $(\mathbf{x}_N^*, \mathbf{t}_N^*)$ is reached. Therefore, the total profit achieved by the prime contractor is $\sum_{i \in N} [\phi_i(t_i^0 - t_i^*) - \Theta_i x_i^*]$, which is the maximum that the prime contractor can obtain by optimal crashing of the tasks in the project. This establishes that Algorithm AE finds a solution which is optimal for the prime contractor.

Finally, we consider the time complexity of Algorithm AE. In Step 1, we solve a linear program with time complexity $O(LP(2n))$, where $LP(2n)$ is the computing time needed to solve the linear program $PC(N)$. The bisection search in Step 2 requires $O(\log(X_U)LP(2n))$ time. Observe that $X_U \leq \sum_{i=1}^n \bar{x}_i$. Step 3 may

require the bisection search of Step 2 to run up to n times, because for each q the maximum crashing time of at least one task is saturated, and there are n tasks in the problem. Step 4 also requires the solution of a linear program with time complexity $O(LP(2n))$. Therefore, the overall time complexity of Algorithm AE is $O(n \log(\sum_i \bar{x}_i)LP(2n))$. ■

In Theorem 4, we assume that the subcontractors crash as long as the bonus and the cost of doing so are balanced; hence, constraint (31) is satisfied at equality. This allows the prime contractor to achieve the maximum benefit $\sum_{i \in N} [\phi_i(t_i^0 - t_i^*) - \Theta_i x_i^*]$ of the system owing to crashing. We note that this is also reasonable in practice for the subcontractors, if the crashing cost includes a minimum acceptable markup to each subcontractor. On the other hand, if the crashing cost does not include such a markup, and if subcontractor i requires at least a share $\underline{\alpha}_i > 0$ of benefit to complete his task earlier, then α_i is replaced by $\alpha_i - \underline{\alpha}_i$ in the left-hand side of (31). In this case, the subcontractors perform all the crashing operations necessary to achieve the solution $(\mathbf{x}_N^*, \mathbf{t}_N^*)$, and the prime contractor's profit becomes $\sum_{i \in N} [(1 - \underline{\alpha}_i) \phi_i(t_i^0 - t_i^*) - \Theta_i x_i^*]$.

4.3 Nonuniform Contracts vs. Uniform Contracts

Since using nonuniform contracts may generate legal and administrative concerns, a question is whether the use of nonuniform contracts is worthwhile for the prime contractor. We conduct a set of computational experiments to address this question. For each instance, we evaluate the cost savings of the prime contractor, the profits of the subcontractors, and the

profits of the overall project, under the two different types of contracts. To focus on the impact of different contracts, we assume that there is only one type of resource. In fact, qualified human resources are the most important and the only type of resources in many projects, such as ITO projects.

Our experimental settings are as follows. The total number of tasks is $n = 25$. The average number of predecessors of each task is 3. The task durations are $d_i \sim U[1, 10]$, i.e. randomly drawn from a uniform distribution in the range $[1, 10]$. The resource upper bounds are $\bar{r}_i \sim U[1, 20]$, $i \in \mathcal{N}$. The resource consumption rates are $r_i \sim U[1, 10]$. The unit completion time costs are $\phi_i \sim U[50, 150]$. The unit crashing costs are $\Theta_i \sim U[40, 120]$. The profit sharing lower bounds are $\underline{\alpha}_i \sim U[\alpha' - 0.2, \alpha' + 0.2]$, where α' is a parameter that we vary in our experiments. In the case of uniform contracts, we let $\underline{\alpha} = \alpha'$. For each value of α' , we generate 1,000 instances and compute the mean values. Our numerical results are summarized in Table 1.

From Table 1, we observe that the cost saving of the overall system remains unchanged as α' increases from 0.2 to 0.8. We also have the following observations.

- *The prime contractor's cost saving decreases as α' increases; however, for each value of α' , the prime contractor's profit with nonuniform contracts is always greater than that with uniform contracts.*

Next, we investigate the performance of the two types of contract as the project becomes larger and more complex. The results are summarized in Table 2.

From the results in Table 2, we have the following observations:

- *As n increases from 25 to 200, the total cost saving % increases slowly, which implies that cooperation results in greater benefit to the overall project as n increases.*

- *The cost saving obtained by the prime contractor also increases when the project contains more tasks.*

- *The relative benefit of nonuniform contracts compared with uniform contracts increases as n increases.*

Finally, we conduct an experiment which varies the average number of predecessors, n^p , of the tasks. As we vary n^p from 2 to 6, the precedence structure of the project becomes more dense. For each value of n^p , we compare the cost savings obtained by the prime contractor under the two different contract strategies. The results are summarized in Table 3, from which we have the following observations:

- *As the precedence structure becomes more dense, the optimal crashing schedule motivated by the prime contractor achieves greater cost savings.*

- *Using nonuniform contracts, the prime contractor gains more of the total cost saving.*

Intuitively, an increase in α' implies that subcontractors receive a higher proportion of benefits from cost-savings of the overall system, resulting in a decrease in cost savings for the prime contractor. Besides, the more complex the project, the more likely the crashing may take place, therefore cooperation among subcontractors can bring more benefits for the project. From the perspective of the prime contractor, however, nonuniform contracts appear to perform better in the numerical experiments. As we surmised before, nonuniform contracts allow the prime contractor to incentivize subcontractors with sufficient resources

Table 1 PC's and SCs' Savings under Uniform vs. Nonuniform Contracts; $n = 25$

α'	Normal cost	Total Saving	PC's Saving (Uniform)	PC's Saving (Nonuniform)	Improve (%)	SCs' Profit (Uniform)	SCs' Profit (Nonuniform)	Improve (%)
0.2	49225	26023	16099	20172	25	9924	5851	-41
0.3	49225	26023	13442	17032	27	12581	8991	-29
0.4	49225	26023	10814	14017	30	15209	12006	-21
0.5	49225	26023	8179	11036	35	17845	14987	-16
0.6	49225	26023	5590	8174	46	20434	17849	-13
0.7	49225	26023	3001	5281	76	23023	20742	-10
0.8	49225	26023	386	2768	617	25637	23255	-9

*Note: SCs' profits exclude the normal fixed payment γ_i .

Table 2 PC's and SCs' Savings under Uniform vs. Nonuniform Contracts; $\alpha' = 0.5$

n	Normal cost	Total Saving	Total Saving / Normal cost (%)	PC's Saving (Uniform)	PC's Saving (Nonuniform)	Improve (%)
25	49225	26023	52.9	8179	11036	35
50	104689	57631	55.0	17708	24960	41
100	208835	115976	55.5	35314	50551	43
150	329119	184626	56.5	55864	80934	45
200	445631	253270	56.8	76473	111299	46

Table 3 PC's and SCs' Savings under Uniform vs. Nonuniform Contracts for Various n^p Values

n^p	Normal cost	Total Saving	Total Saving / Normal cost (%)	PC's Saving (Uniform)	PC's Saving (Nonuniform)	Improve (%)
2	52180	27547	52.8	8724	11860	36
3	104689	57631	55.0	17708	24960	41
4	124405	70976	57.1	21762	31433	44
5	147802	85401	57.8	26337	38612	47
6	165347	99163	60.0	30442	45555	50

or in key positions in a more targeted manner, therefore leading to greater cost savings.

To summarize our computational results, we conclude that uniform and nonuniform contracts generate the same cost saving to the overall system. However, the prime contractor can achieve greater cost saving by using nonuniform contracts. Moreover, the larger the number of tasks and precedence constraints, the larger the profit share that the prime contractor can obtain. Therefore, if the

prime contractor can address any legal or administrative problems that arise from the use of nonuniform contracts, then they are a better choice for her.

5. Sensitivity to Parameter Estimation

In this section, we study the sensitivity of the prime contractor's and subcontractors' profits to the estimation accuracy of two types of parameters: the crashing costs and the resources of the subcontractors. Crashing costs

are typically estimated by the prime contractor, based on cost information commonly available in the marketplace when the outsourcing contracts are established, whereas resources are estimated by subcontractors and submitted to the prime contractor as a part of their tenders. However, these parameters are subject to unpredictable changes during the execution of the project (Tadelis and Bajari 2009). If contracts are designed based on poorly estimated values, then some subcontractors may find that the compensation offered by the prime contractor is insufficient for them to crash the tasks. As a result, the tasks will not be crashed and the overall project profit may be reduced. We study these effects computationally.

5.1 Subcontractors' Unit Crashing Cost

We first evaluate the effects of inaccurate estimate of Θ_N . Our numerical experiment considers nonuniform contracts, and is based on the example in Appendix A. Suppose that the crashing costs estimated at contract formation by the prime contractor are $\tilde{\Theta}_N = (0.5, 1.5, \tilde{\Theta}_3, 2, 1.5)$, where $\tilde{\Theta}_3$ is a parameter which we vary in order to study the sensitivity of project performance to estimation. Suppose that the real crashing costs, as realized during project execution, are $\Theta_N = (1, 1, 3, 1, 1)$. For simplicity, we set $\alpha_i = 0, \forall i \in N$.

We vary the estimated value of $\tilde{\Theta}_3$ from 5 to 1 in 0.5 decrements. For each estimate, we derive the actual crashing amounts used by the subcontractors, and calculate the benefits of the prime contractor and the subcontractors. These results appear in Table 4.

We discuss the results in Table 4. First, if $\tilde{\Theta}_3 > 3$, it is not worthwhile for the prime con-

tractor to crash tasks 3 and 4 together, since $\tilde{\Theta}_3 + \tilde{\Theta}_4 > \phi_5 = 5$. Hence, she does not include the crashing option in the contracts with subcontractors 3 and 4. If $\tilde{\Theta}_3 \leq 3$, then it is worthwhile for the prime contractor to have tasks 3 and 4 crashed. Therefore, she may design contracts that compensate subcontractors 3 and 4 for crashing. The result is that subcontractors 3 and 4 implement the crashing, the prime contractor's cost saving increases accordingly, and the subcontractors also gain some extra profit due to compensation by the prime contractor. When $\tilde{\Theta}_3 < 2$, the compensation of the prime contractor based on this estimate is insufficient to offset the real crashing cost for tasks 3 and 4. Hence, subcontractors 3 and 4 will not implement the crashing as the prime contractor expects, which reduces her profit.

We have the following observations:

- *Either overestimating or underestimating their crashing cost information can reduce the subcontractors' profit.* From Table 4, if subcontractor 3 states his cost to be greater than 3 and the prime contractor accepts this estimate, then the extra profit that the subcontractors can obtain is only 1. The reason is that overestimating the crashing cost may mislead the prime contractor into thinking that it is not worthwhile to crash the task. On the other hand, if subcontractor 3 underestimates his cost, by stating it to be smaller than 2, then the prime contractor provides less compensation. Hence, the subcontractors will find that it is not worthwhile to crash tasks 3 and 4. The results show that the more accurate estimates, e.g., $\tilde{\Theta}_3 = 3$ or $\tilde{\Theta}_3 = 2.5$, allow the subcontractors to

Table 4 Effects of Estimation of Θ_3 on the Prime Contractor's and Subcontractors' Profits

$\tilde{\Theta}_3$	PC's Expected	SCs' Crashing						PC's Actual	SCs' Extra
	Cost Saving	x_1^*	x_2^*	x_3^*	x_4^*	x_5^*	t_5^*	Cost Saving	Profit
5.0	15	1	2	0	0	1	6	15	1
4.5	15	1	2	0	0	1	6	15	1
4.0	15	1	2	0	0	1	6	15	1
3.5	15	1	2	0	0	1	6	15	1
3.0	15	1	2	2	2	1	4	15	3
2.5	16	1	2	2	2	1	4	16	2
2.0	17	1	2	2	2	1	4	17	1
1.5	18	1	2	0	0	1	6	15	1
1.0	19	1	2	0	0	1	6	15	1

gain more.

- *Cooperation among the subcontractors mitigates the effects of inaccurate estimation.* When $2 \leq \tilde{\Theta}_3 \leq 3$ in Table 4, the unit crashing cost of task 3 is underestimated. But the crashing amounts implemented by the subcontractors remain the optimal values. The reason is that, although task 3's unit crashing cost is underestimated, the unit crashing cost of task 4 is overestimated. Through cooperation between the two subcontractors, the effects of these inaccurate estimates offset each other. This result is more pronounced when there are more tasks in the project, which increases the pooling effect.

In Section 4, we observe that the prime contractor always achieves more profit using a nonuniform contract. We now investigate whether this result is also true under inaccurate parameter estimates. We conduct a numerical experiment based on the example of Appendix A to study this issue. We let the unit crashing cost Θ_i , $i \in \mathcal{N}$, follow the uniform distribution $U[40, 120]$, and the unit completion time cost ϕ_i , $i \in \mathcal{N}$, follow the uni-

form distribution $U[50, 150]$. We also let the resource consumption rate r_{i1} , $i \in \mathcal{N}$, follow the uniform distribution $U[1, 10]$ and the resource upper bound $\bar{r}_{i1}$, $i \in \mathcal{N}$, follow the uniform distribution $U[1, 20]$. Furthermore, we let α_i , $i \in \mathcal{N}$, follow the uniform distribution $U[\alpha' - 0.2, \alpha' + 0.2]$. We fix $\alpha' = 0.5$, and let $\underline{\alpha} = \alpha'$ in the case of uniform contracts. We assume that all estimates are accurate except $\tilde{\Theta}_3$. We vary the estimate of $\tilde{\Theta}_3$ from 50% to 200% of the real value Θ_3 , and evaluate how the profits of the overall project, the prime contractor and the subcontractors change. The results are reported in Table 5.

From the results in Table 5, we have the following observations.

- *The total saving of the project under nonuniform contracts is less than that under uniform contracts.* The reason is that the greater sharing of crashing cost with subcontractors under uniform contracts compensates for the poor estimation of the crashing cost. This result again shows that the pooling effect resulting from cooperation offsets the effects of inaccurate estimation.

Table 5 Effects of Contract Type on Profits

$\tilde{\Theta}_3$ vs Θ_3	Expected Total Saving (*)	Real Total Saving (Uni)	Diff from (*) (%)	Real PC's Saving (Uni)	Real SCs' Profit (Uni)	Real Total Saving (Non)	Diff from (*) (%)	Real PC's Saving (Non)	Impr vs Uni (%)	Real SCs' Profit (Non)	Impr vs Uni (%)
0.50	601.6	552	-8.2	299.8	252.2	541.2	-10.6	439	46.4	102.2	-59.5
0.67	579	552	-4.4	298.5	253.5	541.2	-6.8	439	47	102.2	-59.7
0.75	571.1	552	-3.1	297.8	254.2	541.2	-5.6	439	47.4	102.2	-59.8
0.80	567.9	554.4	-2.1	299.7	254.8	541.2	-5.1	439	46.5	102.2	-59.9
1.00	556.4	556.4	0	298.1	258.3	556.4	0	454.1	52.4	102.2	-60.4
1.25	548.3	556.4	1.6	290.5	265.9	554.4	1.4	446	53.5	108.4	-59.2
1.33	546.9	554.4	1.8	292.8	261.7	552	1.3	444.6	51.9	107.4	-59
1.50	544.2	552	2.1	291.7	260.3	552	2.1	442	51.5	110	-57.7
2.00	541.2	546.4	2.1	289.1	257.3	541.2	0	439	51.9	102.2	-60.3

We complete this section with an experiment where the crashing costs of all the subcontractors, instead of only one of them, are inaccurately estimated. We use the same data as in the experiment for Table 5, except that we vary all the ratios $\frac{\tilde{\Theta}_i}{\Theta_i}$, $\forall i \in \mathcal{N}$, together from 50% to 200%. Our results are summarized in Table 6.

We compare the results in Table 6 with those in Table 5. We observe that:

- *It is better for the prime contractor to overestimate the crashing costs of the subcontractors than to underestimate them.* The main reason is that a contract with underestimated crashing costs can make a subcontractor reluctant to perform crashing, hence the optimal crashing expected by the prime contractor does not take place. Then, the most efficient solution for the prime contractor is not achieved, and the resulting loss is typically greater than in the case of overestimation.
- *The use of nonuniform contracts is a more robust strategy for the prime contractor than the*

use of uniform contracts. Although a uniform contract provides greater cost savings to the prime contractor in underestimation cases, the improvements there are typically less than 10%. Whereas, in overestimation cases, the benefits from nonuniform contracts are typically greater than 60%.

5.2 Subcontractors' Resource Availability

The available resources are specified in the subcontractors' tenders. However, this information is subject to unpredictable changes during project execution. For example, the available labor or materials may change due to uncontrollable events, such as a strike or a supply chain disruption. Therefore, the prime contractor may also need to estimate this information when she designs the outsourcing contracts.

However, from Theorem 1, the subcontractors' cooperative game is balanced. Hence, they pool their resources within the grand

Table 6 Project Members' Payoffs under Uniform vs. Nonuniform Strategies

$\tilde{\Theta}_i$ vs Θ_i	Expected Total Saving (*)	Real Total Saving (Uni)	Diff from (*) (%)	Real PC's Saving (Uni)	Real SCs' Profit (Uni)	Real Total Saving (Non)	Diff from (*) (%)	Real PC's Saving (Non)	Impr vs Uni (%)	Real SCs' Profit (Non)	Impr vs Uni (%)
0.50	838.3	592.7	-32.4	389.3	203.4	461.3	-48.5	361.5	-7.1	99.9	-50.9
0.67	765.6	605.9	-23.7	378.3	227.6	461.3	-43.5	361.5	-4.5	99.9	-56.1
0.75	726.8	614.1	-17.8	371.6	242.5	461.3	-40.4	361.5	-2.7	99.9	-58.8
0.80	705.8	619.4	-14.2	368	251.4	461.3	-38.5	361.5	-1.8	99.9	-60.3
1.00	624.8	624.8	0	334.2	290.5	624.8	0	524.9	57	99.9	-65.6
1.25	534.7	621.9	22.4	283.9	337.9	616.7	20.4	464.6	63.6	152.1	-55
1.33	508.8	614.8	29.6	271.4	343.5	610.6	25.4	447	64.7	163.6	-52.4
1.50	458.6	607.6	46.6	243.8	363.8	595.5	37	410.5	68.4	185.1	-49.1
2.00	337.3	586.3	111	173.1	413.2	550.1	81.2	315.9	82.5	234.3	-43.3

coalition, which makes it unnecessary for the prime contractor to estimate the resources of each individual subcontractor. Therefore, the prime contractor only needs to estimate the total amount of resources. Let $\tilde{r}$ denote the prime contractor's estimate of the total amount, and $\bar{r}$ denote the actual amount. We use a computational experiment to study the impact of $\tilde{r}$ on the profits of the prime contractor and the subcontractors.

The following experiment considers nonuniform contracts, and is based on the example of Appendix A, with the following modifications. Assume $\underline{\alpha}_i = 0, \forall i \in \mathcal{N}$. The real unit crashing costs are $\Theta_{\mathcal{N}} = (1, 1, 2, 1, 1)$, but the estimated values are $\tilde{\Theta}_{\mathcal{N}} = (2, 1.5, 2.5, 2, 1.5)$. The resources of the subcontractors are $\bar{r}_{\mathcal{N}} = (1, 1, 1, 1, 2)$, with a total of 6. We assume that the prime contractor estimates the total resource amount. We vary this estimate, $\tilde{r}$, from 3 to 9. For each value of $\tilde{r}$, we derive the actual schedule used by the subcontractors, the maximum variable profit which the subcontractors receive, and

the cost saving of the prime contractor. The results are summarized in Table 7.

From the results in Table 7, we have the following observations:

- *Both underestimation and overestimation of the total available resources may reduce the prime contractor's profit.* From Table 7 we observe that, when the prime contractor underestimates the subcontractors' resources, her profit is reduced, although the project is implemented as expected, with crashing at the upper bounds specified in the contract. However, when the prime contractor overestimates the amount of resources, the project may not be implemented as expected, and her profit is again reduced.
- *The subcontractors' profit increases as the prime contractor's estimate of the total available resources increases.* The increase as $\tilde{r}$ changes from 3 to 6 occurs because the prime contractor also sets the value of $\bar{x}_{\mathcal{N}}$ larger, hence the subcontractors perform more task crashing and receive

Table 7 Effect of Total Resource Estimation on Profits

$\tilde{r}$	PC's Expected	SCs' Schedule						PC's Actual	SCs' Extra
	Cost Saving	x_1^*	x_2^*	x_3^*	x_4^*	x_5^*	t_5^*	Cost Saving	Profit
3	10	1	1	0	0	1	7	10	2
4	13.5	1	2	0	0	1	6	13.5	2.5
5	13.5	1	2	0	0	1	6	13.5	2.5
6	14	1	2	1	1	1	5	14	4
7	14	1	2	1	1	1	5	14	4
8	14.5	1	1	2	1	1	5	$12\frac{1}{3}$	$4\frac{2}{3}$
9	14.5	1	1	2	1	1	5	$12\frac{1}{3}$	$4\frac{2}{3}$

more compensation from the prime contractor. However, the increase in profit as $\tilde{r}$ changes from 6 to 9 has a different explanation. Since the prime contractor overestimates the crashing cost of tasks 3 and 4, the subcontractors crash task 3 instead of task 2, which increases their profit.

Finally, we examine the effect of contract type on the profits of the various project members. Our study is based on the parameter settings of the example in Table 7. Suppose that the prime contractor estimates the total resource of the subcontractors. For each $i \in \mathcal{N}$, $\frac{\tilde{r}_i}{\bar{r}_i}$ represents the ratio of the estimated total resource to the actual total. We let this ratio follow the distribution $U[\bar{r} - 0.2, \bar{r} + 0.2]$, where $\bar{r}$ represents the mean value. In this experiment, we vary $\bar{r}$ from 0.5 to 2, which implies that the prime contractor's estimation varies from 50% less than the correct value to 100% more. We study the effect of contract type on profits. The results are summarized in Table 8.

From the results in Table 8, we have the following observations:

- Under both contract types, the subcontractors crash as the prime contractor expects, unless there are not enough resources. Hence, the

different parameter settings in these two contracts do not affect the optimal schedule used by the subcontractors; they only change the profit allocation between the prime contractor and the subcontractors. Under nonuniform contracts, the prime contractor receives greater cost savings. The improvement over uniform contracts is about 85%, and does not vary significantly with $\bar{r}$.

- Once again, a pooling effect from the subcontractors' cooperation mitigates the loss of profit due to inaccurate estimation. Although we vary $\bar{r}$ from 0.5 to 2.0, the difference between the expected and actual savings is relatively small. The reason is that the prime contractor's decision making problem is only affected by the total resources of all the subcontractors.

6. Concluding Remarks

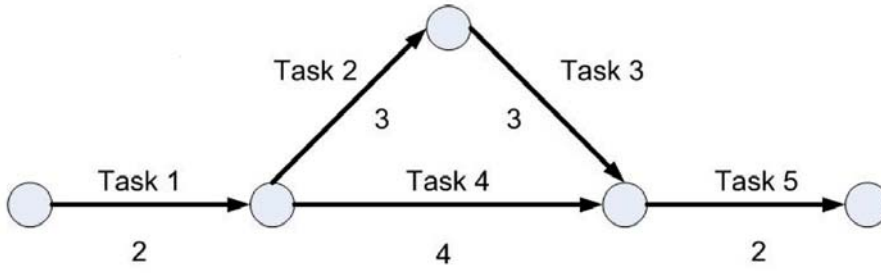
In this paper, we study cooperation and contract design issues involving a prime contractor and several subcontractors who perform the tasks outsourced by the prime contractor. The prime contractor receives from the project owner an amount of revenue which is a non-decreasing function of the completion times

Table 8 Effects of Contract Type on Profits

$\dot{r}$	Expected Total Saving (*)	Real Total Saving (Uni)	Diff from (*) (%)	Real PC's Saving (Uni)	Real SCs' Profit (Uni)	Real Total Saving (Non)	Diff from (*) (%)	Real PC's Saving (Non)	Impr vs (%)	Real SCs' Profit (Non)	Impr vs (%)
0.50	1010.76	1010.76	0	370.84	639.9	1010.76	0	680.66	83.86	330.1	-48.48
0.67	1098.18	1098.18	0	402.52	695.68	1098.18	0	744.14	84.94	354.08	-49.04
0.75	1131.2	1131.2	0	414.42	716.78	1131.2	0	768.66	85.44	362.56	-49.32
0.80	1144.34	1144.34	0	419.26	725.12	1144.34	0	778.28	85.56	366.06	-49.38
1.00	1177.18	1177.18	0	431.04	746.12	1177.18	0	801.96	85.74	375.22	-49.5
1.25	1195.56	1177.18	-1.44	431.04	746.12	1177.18	-1.44	802.2	85.82	374.98	-49.54
1.33	1198.54	1177.18	-1.62	431.04	746.12	1177.18	-1.62	802.26	85.82	374.92	-49.54
1.50	1205.2	1177.18	-2.02	431.04	746.12	1177.18	-2.02	802.16	85.8	375	-49.52
2.00	1216.08	1177.18	-2.56	431.04	746.12	1177.18	-2.56	802.22	85.82	374.94	-49.54

of the tasks. To increase her profit, she provides contractual incentives to the subcontractors to crash their tasks and deliver them earlier as required. The subcontractors can cooperate so as to maximize their profits, and we model this decision as a cooperative game. We show that this game, although not convex, is balanced. Hence, the subcontractors can always cooperate, and we describe a stable profit sharing scheme. We extend these results to allow task lengthening, by which some subcontractors deliberately execute their tasks at a slower speed, so that resources can be released to other subcontractors. We consider the prime contractor's contract design problem using uniform contracts for all the subcontractors, and also nonuniform ones that are customized for different subcontractors. We provide efficient algorithms for computing the optimal contract parameters in both cases. We also conduct computational experiments to evaluate the sensitivity of the profits of the project members to the accuracy of parameter estimation.

Our work has several implications for the practice of project management. A positive result is that cooperation among the subcontractors can be achieved. This facilitates efficient execution of the project, and therefore provides a better solution for all the parties involved, including the subcontractors and the prime contractor. Our work also shows the value of obtaining accurate information about the cost and resource parameters for crashing the tasks. As is typical in practice, the crashing costs and resource consumption rates are common information in the marketplace. Also, the available resources are a commitment of the subcontractors in their original tenders. These parameters are, however, subject to unpredictable changes during project execution. It is valuable to observe, from our computational experiments, that the pooling effect due to the subcontractors' cooperation mitigates the likely loss from poor parameter estimation. Also, we find that there are situations where the subcontractors may lose by submitting inaccurate information, since doing so reduces the efficiency of

**Figure 3** Project Network**Table 9** Data for the Example Project

Parameters	Task 1	Task 2	Task 3	Task 4	Task 5
d_i	2	3	3	4	2
$\bar{r}_{i1}$	1	1	1	1	4
r_{i1}	1	1	1	1	1
Θ_i	1	1	2	1	1

the overall project and accordingly their own profit. If the prime contractor can make this point clear to the subcontractors, then they are more likely to submit information that is as accurate as possible, thus supporting optimization of the overall system from the prime contractor's perspective. Exploring project management with outsourcing also has important implications for decision makers of emerging platforms that connect the job supply side and job processing side.

There are several promising directions for related future research. First, although we are able to identify an allocation in the core using the optimal dual solution, the problem of checking whether a given profit distribution is in the core or not remains to be studied. Solving this problem would facilitate negotiations among the subcontractors. Second, while we study a project with linear crashing costs, it would be valuable to consider more general costs. A practical example is piecewise linear and convex crashing costs. Finally, our

work can be extended to consider issues of resource sharing and cooperation across multiple projects. In conclusion, we hope that our work will motivate the study of these important research issues that lie at the intersection of project management, cooperative game theory, and contract design.

Appendix A: An Example Project

Consider a project with 5 tasks, as shown in Figure 3. We assume there is only one type of resource, i.e., $\mathcal{R} = \{1\}$. The project parameters are shown in Table 9.

Under the normal schedule, the project is implemented with the task times shown in Figure 3, and the completion time of the project is 10. Suppose that the prime contractor can save $5x$ dollars if the project is finished $x \geq 0$ time units earlier, i.e. $\phi_N = (0, 0, 0, 0, 5)^T$. Assume that the prime contractor signs outsourcing contracts with the subcontractors with parameters as shown

Table 10 Outsourcing Contract Data

Variables	Task 1	Task 2	Task 3	Task 4	Task 5
φ_i	0	0	0	0	5
ϑ_i	0	0	0	0	0
t_i^0	2	5	8	8	10
$\underline{t}_i$	1	2	3	3	4
$\bar{x}_i$	1	2	2	2	1

in Table 10.

Then, problem $SC(\mathcal{N})$ is

$$\max v(\mathcal{N}) = -x_1 - x_2 - 2x_3 - x_4 - x_5 - 5t_5 + \sum_{i=1}^5 \gamma_i + 50 \quad (34)$$

s.t.

$$\begin{aligned} & \begin{pmatrix} 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 \end{pmatrix} \mathbf{x}_{\mathcal{N}} + \begin{pmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 \end{pmatrix} \mathbf{t}_{\mathcal{N}} \leq \begin{pmatrix} -3 \\ -3 \\ -4 \\ -2 \\ -2 \end{pmatrix} \\ & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \mathbf{x}_{\mathcal{N}} + \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \mathbf{t}_{\mathcal{N}} \leq \begin{pmatrix} 2 \\ -2 \\ 10 \end{pmatrix} \\ & \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \end{pmatrix} \mathbf{x}_{\mathcal{N}} \leq 8 \\ & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \mathbf{x}_{\mathcal{N}} \leq \begin{pmatrix} 1 \\ 2 \\ 2 \\ 2 \\ 1 \end{pmatrix} \\ & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \mathbf{t}_{\mathcal{N}} \geq \begin{pmatrix} 1 \\ 2 \\ 3 \\ 3 \\ 4 \end{pmatrix} \\ & \mathbf{x}_{\mathcal{N}}, \mathbf{t}_{\mathcal{N}} \geq 0 \end{aligned}$$

Consider the coalition $\mathcal{S} = \{2, 3, 4\}$. Problem $SC(\mathcal{S})$ can be written as follows:

$$\max v(\mathcal{S}) = -x_2 - 2x_3 - x_4 + \gamma_2 + \gamma_3 + \gamma_4 \quad (35)$$

s.t.

$$\begin{aligned}
 & \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \mathbf{x}_S + \begin{pmatrix} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{t}_S \leq \begin{pmatrix} -5 \\ -3 \\ -6 \\ 8 \\ 8 \end{pmatrix} \\
 & \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \mathbf{x}_S \leq 3 \\
 & \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{x}_S \leq \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \\
 & \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{t}_S \geq \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} \\
 & \mathbf{x}_S, \mathbf{t}_S \geq 0
 \end{aligned}$$

The matrices $\mathbf{A}_S^i$, $i = 1, 2$, and the vector $\mathbf{b}_S^1$ from (8) are shown in the first constraint above. Note that $\mathbf{B}_S^i$, $i = 1, 2$, and $\mathbf{b}_S^2$ from (9) are all zero, since constraint (9) does not exist for coalition S in this example.

Appendix B: Dual of the Example Problem

Consider the example introduced in Appendix A. The dual problem $\text{SD}(\mathcal{N})$ is:

$$\begin{aligned}
 \min \quad & (-3, -3, -4, -2, -2)(\rho_{21}, \rho_{32}, \rho_{41}, \rho_{53}, \rho_{54})^\top + (2, -2, 10)\tau_N + 8\omega_1 + (1, 2, 2, 2, 1)\mu_N \\
 & -(1, 2, 3, 3, 4)\zeta_N + \sum_{i=1}^5 \gamma_i + 50
 \end{aligned}$$

s.t.

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 1 & -1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 1 & 0 & 1 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} \rho_N \\ \tau_N \\ \omega_1 \\ \mu_N \\ \zeta_N \end{pmatrix} \geq \begin{pmatrix} -1 \\ -1 \\ -2 \\ -1 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -5 \end{pmatrix}$$

$$\mathbf{z}_N \geq 0$$

The dual problem of coalition $\mathcal{S}$, $\text{SD}(\mathcal{S})$, can be written as follows:

$$\min \quad (-5, -3, -6, 8, 8)(\rho_{21}, \rho_{32}, \rho_{41}, \rho_{53}, \rho_{54})^\top + 3\omega_1 + (2, 2, 2)\mu_{\mathcal{S}} - (2, 3, 3)\zeta_{\mathcal{S}} + \sum_{i=2}^4 \gamma_i$$

s.t.

$$\begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} \rho_{\mathcal{S}} \\ \omega_1 \\ \mu_{\mathcal{S}} \\ \zeta_{\mathcal{S}} \end{pmatrix} \geq \begin{pmatrix} -1 \\ -2 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\mathbf{z}_{\mathcal{S}} \geq \mathbf{0}$$

In order to consider $\mathbf{z}_{\mathcal{S}}$ and $\mathbf{z}_{\mathcal{N}}$ with the same dimension, we rewrite $\text{SD}(\mathcal{S})$ as follows:

$$\begin{aligned} \min \quad & (-5, -3, -6, 8, 8)(\rho_{21}, \rho_{32}, \rho_{41}, \rho_{53}, \rho_{54})^\top + (0, 0, 0)\tau_{\mathcal{S}} + 3\omega_1 + (0, 2, 2, 2, 0)\mu_{\mathcal{S}} \\ & - (0, 2, 3, 3, 0)\zeta_{\mathcal{S}} + \sum_{i=2}^4 \gamma_i \end{aligned}$$

s.t.

$$\begin{pmatrix} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} \rho_{\mathcal{S}} \\ \tau_{\mathcal{S}} \\ \omega_1 \\ \mu_{\mathcal{S}} \\ \zeta_{\mathcal{S}} \end{pmatrix} \geq \begin{pmatrix} -1 \\ -2 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\mathbf{z}_{\mathcal{S}} \geq \mathbf{0}$$

The matrix in problem $\text{SD}(\mathcal{S})$ consists of rows 2,3,4,7,8 and 9 of the matrix in problem $\text{SD}(\mathcal{N})$.

Appendix C: Payoff Distribution for the Example

Consider the example in Appendix A. Given the contracts with the prime contractor, the grand coalition's maximum profit can be obtained by solving problem $\text{SC}(\mathcal{N})$. An optimal solution is

$$\mathbf{z}_{\mathcal{N}}^* = (2, 2, 1, 2, 1, 0, 2, 0, 0, 0, 1, 0, 0, 2, 2, 0, 0, 0, 2)$$

That is, $\rho_{21}^* = 2, \rho_{32}^* = 2, \rho_{41}^* = 1, \rho_{53}^* = 2, \rho_{54}^* = 1, \tau_2^* = 1, \mu_2^* = 1, \mu_5^* = 2, \zeta_1^* = 2, \zeta_5^* = 2$, with all other variables equal to zero. Hence, the optimal value is $v(\mathcal{N}) = \sum_{i=1}^5 \gamma_i + 20$. From Theorem 1, we have

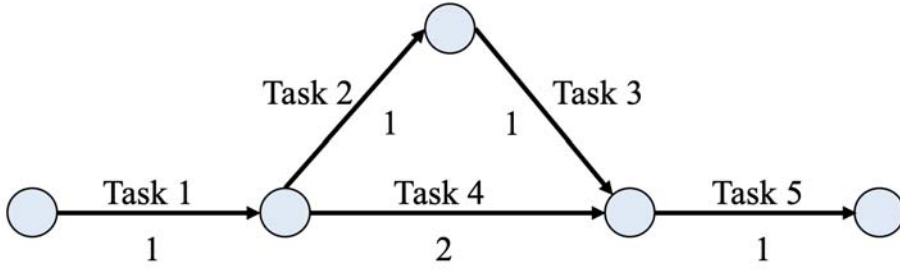


Figure 4 Crashing Schedule

$$\begin{cases} \lambda_1 = \gamma_1 - \zeta_1^* + 2 * (\rho_{21}^* + \rho_{41}^*) - 2 * \tau_2^* = \gamma_1 + 2 \\ \lambda_2 = \gamma_2 + 2 * \mu_2^* + 5 * \rho_{32}^* - (3 + 2)\rho_{21}^* = \gamma_2 + 2 \\ \lambda_3 = \gamma_3 + 8\rho_{53}^* - (3 + 5)\rho_{32}^* = \gamma_3 \\ \lambda_4 = \gamma_4 + 8\rho_{54}^* - (4 + 2)\rho_{41}^* = \gamma_4 + 2 \\ \lambda_5 = \gamma_5 + 5 * t_5^0 - 4 * \zeta_5^* + \mu_5^* - (2 + 8)\rho_{53}^* - (2 + 8)\rho_{54}^* = \gamma_5 + 14 \end{cases}$$

and the new schedule is shown in Figure 4.

Without cooperation, the subcontractors' profits are $(\gamma_1, \gamma_2, \gamma_3, \gamma_4, \gamma_5)$. With cooperation and after distribution from the grand coalition, their profits become $(\gamma_1 + 2, \gamma_2 + 2, \gamma_3, \gamma_4 + 2, \gamma_5 + 14)$, respectively.

Appendix D: Task Lengthening in the Example

We again consider the example in Appendix A. Let $\bar{r}_i = 0, i = 1, \dots, 5$, which implies that no

subcontractors own any resources for crashing. Thus, in order to perform crashing, some tasks need to be lengthened in order to release resources.

Solving problem $SC(\mathcal{N})$, we obtain the optimal solution: $t_1^* = 1, t_2^* = 4, t_3^* = 7, t_4^* = 7, t_5^* = 8$, which implies that task 4 is lengthened by 2 time units to release 2 units of the resource, task 1 is crashed by 1 time unit and task 5 is crashed by 1 time unit.

By solving the problem $SD(\mathcal{N})$, we obtain the optimal dual solution:

$$\rho_{21} = 2.5, \rho_{32} = 2.5, \rho_{41} = 2.5, \rho_{53} = 2.5, \rho_{54} = 2.5, \tau_2 = 3.5, \omega_1 = 2.5, \mu_5 = 1.5, \zeta_1 = 1.5$$

while other variable values are 0. Using (24), we compute the core distribution:

$$\begin{cases} \lambda_1 = \gamma_1 - \zeta_1 - d_1\tau_2 + t_1^0\rho_{21} + t_1^0\rho_{41} = \gamma_1 - 1.5 - 2 * 3.5 + 2 * 2.5 + 2 * 2.5 = \gamma_1 + 1.5 \\ \lambda_2 = \gamma_2 - (t_1^0 + d_2)\rho_{21} + t_2^0\rho_{32} = \gamma_2 - (2 + 3) * 2.5 - 5 * 2.5 = \gamma_2 \\ \lambda_3 = \gamma_3 - (t_2^0 + d_3)\rho_{32} + t_3^0\rho_{53} = \gamma_3 - (5 + 3) * 2.5 - 8 * 2.5 = \gamma_3 \\ \lambda_4 = \gamma_4 - (t_1^0 + d_4)\rho_{41} + t_4^0\rho_{32} = \gamma_4 - (2 + 4) * 2.5 + 8 * 2.5 = \gamma_4 + 5 \\ \lambda_5 = \gamma_5 + 5t_5^0 + \bar{x}_5\mu_5 - (t_3^0 + d_5)\rho_{53} - (t_4^0 + d_5)\rho_{54} = \gamma_5 + 50 + 1.5 - 10 * 2.5 - 10 * 2.5 = \gamma_5 + 1.5 \end{cases}$$

The above distribution indicates that task 4 is awarded an extra \$5 for the resources which it releases, while task 1 and task 5 are awarded an extra \$1.5 each.

Acknowledgments

The work of the first author is supported by Leading Talent Program of Guangdong Province (Project No. 2016LJ06D703), the National Natural Science Foundation of China (Project No. 72192805), and the Shenzhen Science and Technology Innovation Commission (Project No. JCYJ20210324115604012). The work of the second author is supported by the National Natural Science Foundation of China (Key Research Grant 71732003) and the Summer Fellowship Program of the Fisher College of Business, The Ohio State University. Helpful comments on an earlier draft were provided by Prof. John Gray, The Ohio State University, and Prof. Minghui Lai, Southeast University, China. Finally, we would like to express our sincere gratitude to the Editor and the two referees for their constructive reviews to help improve our work.

Data Availability

All data generated or analysed during this study are included in this published article (and its supplementary information files).

Endnotes

¹ Source: <https://www.statista.com/outlook/tmo/it-services/it-outsourcing/worldwide>

References

- Abu-Hilje SF, Ibbs CW (1989). Schedule-based construction incentives. *Journal of Construction Engineering and Management* 115: 430-443.
- Albino V, Carbonara N, Giannoccaro I (2007). Supply chain cooperation in industrial districts: A simulation analysis. *European Journal of Operational Research* 177: 261-280.
- Amaral J, Anderson E, Parker G (2011). Putting it together: How to succeed in distributed product development. *MIT Sloan Management Review* 52(2): 51.
- Asgari S, Afshar A, Madani K (2014). Cooperative game theoretic framework for joint resource management in construction. *Journal of Construction Engineering and Management* 140: 04013066.
- Bazaraa MS, Jarvis JJ, Sherali HD (1990). *Linear Programming and Network Flows*. John Wiley, New York.
- Bergantiños G, Sánchez E (2002). How to distribute costs associated with a delayed project. *Annals of Operations Research* 109: 159-174.
- Brânzei R, Ferrari G, Fragnelli V, Tijs S (2002). Two approaches to the problem of sharing delay costs in joint projects. *Annals of Operations Research* 109: 359-374.
- Brucker P, Drexel A, Möhring RH, Neumann K, Pesch E (1999). Resource-constrained project scheduling: Notation, classification, models, and methods. *European Journal of Operational Research* 112: 3-41.
- Castro J, Gómez D, Tejada J (2007). A project game for PERT networks. *Operations Research Letters* 35: 791-798.
- Castro J, Gómez D, Tejada J (2008). A rule for slack allocation proportional to the duration in a PERT network. *European Journal of Operational Research* 187: 556-570.
- Chen B, Hall NG (2021). Incentive schemes for resolving Parkinson's Law in project management. *European Journal of Operational Research* 288(2): 666-681.
- Cheung SO, Ng TS, Wong SP, Suen HC (2003). Behavioral aspects in construction partnering. *International Journal of Project Management* 21: 333-343.
- Das TK, Teng BS (1998). Between trust and control: Developing confidence in partner cooperation in alliances. *Academy of Management Review* 23: 491-512.
- Dreesen T, Linden R, Meures C, Schmidt N, Rosenkranz C (2016). Beyond the border: A comparative literature review on communication practices for agile global outsourced software development projects. 2016 49th Hawaii International Conference on System Sciences (HICSS), IEEE. 4932-4941.
- Eriksson PE (2008). Procurement effects on competition in client-contractor relationships. *Journal of Construction Engineering and Management* 134: 103-111.

- Errasti A, Beach R, Oyarbide A, Santos J (2007). A process for developing partnerships with subcontractors in the construction industry: An empirical study. *International Journal of Project Management* 25: 250-256.
- Estévez-Fernández A, Born P, Hamers H (2005). Project games. Working paper, Center and Department of Econometrics and Operations Research, Tilburg University, Tilburg, Netherlands.
- Garib R (2009). Outsourcing: The emerging trend. *Network World*. August 25th. Available at: <http://www.networkworld.com/news/2009/082509-outsourcingthe-emerging.html>.
- Global Projects Network (2013). Available at: <http://www.globalprojectsnetwork.com/>.
- Global Quality & Services (2013). Available at: <http://www.global-qs.com/services/outsourcing-project-staffing.html>.
- Granot D (1986). A generalized linear production model: A unifying model. *Mathematical Programming* 34: 212-222.
- Hafezalkotob A, Hoseinpour E, Damghani K K (2017). A game theory approach for competition and cooperation among project's subcontractors with interaction activities. *Journal of Project Management* 2: 71-86.
- Hall N G (2016). Research and teaching opportunities in project management. *INFORMS*: 329-388. <https://doi.org/10.1287/educ.2016.0146>.
- Hartmann A, Ling F Y Y, Tan J S (2009). Relative importance of subcontractor selection criteria: Evidence from Singapore. *Journal of Construction Engineering and Management* 135: 826-832.
- Hedjazi D, Layachi F, Boubiche D E (2019). A multi-agent system for distributed maintenance scheduling. *Computers & Electrical Engineering* 77: 1-11.
- Herroelen W, De Reyck B, Demeulemeester E (1998). Resource-constrained project scheduling: A survey of recent developments. *Computers & Operations Research* 25: 279-302.
- Hyun J H (1994). Buyer-supplier relations in the European automobile component industry. *Long Range Planning* 27(2): 66-75.
- Jalali Sohi A, Bosch-Rekveltdt M, Hertogh M (2020). Four stages of making project management flexible: Insight, importance, implementation and improvement. *Organization, Technology & Management in Construction: An International Journal* 12: 2117-2136.
- Kumaraswamy M M, Matthews J D (2000). Improved subcontractor selection employing partnering principles. *Journal of Management in Engineering* 16: 47-57.
- Li F, Xu Z, Li H (2021). A multi-agent based cooperative approach to decentralized multi-project scheduling and resource allocation. *Computers & Industrial Engineering* 151: 106961.
- Matthews O, Howell G A (2005). Integrated project delivery an example of relational contracting. *Lean Construction Journal* 2: 46-61.
- Moradi M, Hafezalkotob A, Ghezavati V (2018). Sustainability in fuzzy resource constraint project scheduling in a cooperative environment under uncertainty: Iran's Chitgar Lake case study. *Journal of Intelligent & Fuzzy Systems* 35: 6255-6267.
- Moradi M, Hafezalkotob A, Ghezavati V (2019). Robust resource-constrained project scheduling problem of the project's subcontractors in a cooperative environment under uncertainty: Social complex construction case study. *Computers & Industrial Engineering* 133: 19-28.
- 1000ventures (2012). Customer intimacy: Achieving better results for supplier and customer alike. Available at: http://www.1000ventures.com/business_guide/crosscuttings/customer_intimacy.html.
- Owen G (1975). On the core of linear production games. *Mathematical Programming* 9: 358-370.
- Peleg B, Sudhölter P (2003). *Introduction to the Theory of Cooperative Games*. Kluwer, Boston, MA.
- Perng Y H, Chen S J, Lu H J (2005). Potential benefits for collaborating formwork subcontractors based on cooperative game theory. *Building and Environment* 40: 239-244.
- Pisano G P, Verganti R (2008). Which kind of collaboration is right for you? *Harvard Business Review* 86: 78-86.
- Plaskoff M (2013). How to set a construction schedule and pay schedule. Video available at: [http://www.youtube.com/watch?v=7wicCaCfc\\$_\\$8](http://www.youtube.com/watch?v=7wicCaCfc$_$8).
- Prahalad C K, Hamel G (1990). The core competence of the corporation. *Harvard Business Review* 68: 79-91.
- Quinn J B, Hilmer F G (1994). Strategic outsourcing. *Sloan Management Review* 35: 43-55.
- Samet D, Zemel E (1984). On the core and dual set of linear programming games. *Mathematics of Operations Research* 9: 309-316.

- San Diego Business Journal (2013). Isis Pharmaceuticals receives \$8M in milestone payments. Available at: <http://www.sdbj.com/news/2013/jun/11/isis-pharmaceuticals-receives-8m-milestone-payment/>.
- Shapley L (1953). A value for n -person games. *Annals of Mathematical Studies* 28: 307-317.
- She B, Chen B, Hall N G (2021). Buffer sizing in critical chain project management by network decomposition. *Omega* 102: 102382.
- Shtub A, Etgar R (1997). A branch and bound algorithm for scheduling projects to maximize net present value: The case of time dependent, contingent cash flows. *International Journal of Production Research* 35: 3367-3378.
- Smart Mobility Management (2013). Collaboration between suppliers the key to success. Available at: [http://www.smart-mobilitymanagement.com/collaboration\\$__\\$between\\$__\\$suppliers\\$__\\$the\\$__\\$key\\$__\\$to\\$__\\$success\\$__\\$53773-en-424-184145.html](http://www.smart-mobilitymanagement.com/collaboration$__$between$__$suppliers$__$the$__$key$__$to$__$success$__$53773-en-424-184145.html).
- Sturgeon T J (2002). Modular production networks: A new American model of industrial organization. *Industrial and Corporate Change* 11: 451-496.
- Tadelis S, Bajari P (2009). Incentives and award procedures: Competitive tendering vs. negotiations in procurement. *Handbook of Procurement*, N Dimitri, G Pigo, G Spagnolo(eds): 121-140. Cambridge University Press, published online.
- Teng Y, Li X, Wu P, Wang X Y (2017). Using cooperative game theory to determine profit distribution in IPD projects. *International Journal of Construction Management*. <https://doi.org/10.1080/15623599.2017.1358075>.
- Tully S (1994). You'll never guess who really makes. *Fortune* 130: 124-129.
- U.S. Department of Transportation (2013). Alternative payment and progress reporting methods, task #2. Available at: <http://www.fhwa.dot.gov/programadmin/contracts/etgpayment.cfm>.
- ValueBasedManagement.net (2012). Product life cycle - industry maturity stages. Available at: http://www.valuebasedmanagement.net/methods_product_life_cycle.html.
- Vanhoucke M, Demeulemeester E, Herroelen W (2001). On maximizing the net present value of a project under renewable resource constraints. *Operations Research* 47: 1113-1121.
- Wall Street Journal (2013). Sucampo receives \$10 million milestone payment from Takeda on first sale of AMITIZA(R) (lubiprostone) for OIC in the US. Available at: <http://online.wsj.com/article/PR-CO-20130626-910140.html>.
- Yu X, Petter S (2014). Understanding agile software development practices using shared mental models theory. *Information and Software Technology* 56: 911-921.
- Zuo X D, Xue S J, Liang Y, Luo L (2000). A way to find all the optimal solutions in linear programming. *Journal of Systems Science and Systems Engineering* 9(4): 482-487.

Xiaoqiang Cai is Associate Vice President and Presidential Chair Professor, The Chinese University of Hong Kong (CUHK), Shenzhen. He was Professor and Department Chairman in the Shatin campus of CUHK before joining CUHK-Shenzhen. He holds a PhD in Systems Engineering from Tsinghua University. His research interests are mainly in scheduling models and algorithms, cooperative games and applications, and logistics and supply chain management. He has published over 300 papers in academic journals, books, and conferences, including more than 100 papers in leading journals, such as Operations Research, Management Science, SIAM Journal on Control and Optimization, Production and Operations Management, Transportation Science, and Transportation Research. He is a recipient of the Outstanding Young Scientist award (overseas category) from the National Natural Science Foundation of China, and the IEOM Award of Outstanding Professor in Logistics and Supply Chain. He is Academician of the International Academy for Systems and Cybernetic Sciences, and Fellow of Asia-Pacific Artificial Intelligence Association.

Nicholas G. Hall is Berry Professor of Operations & Business Analytics, Fisher College of Business, The Ohio State University. He holds a Ph.D. in management science (University of California, Berkeley). His research interests are in project management, scheduling, and sports. He has published over 90 articles in Operations Research, Management Science, Mathematics of Operations Research, Mathematical Programming, Games and Economic Behavior, Interfaces, and others. His main teaching interests are in project management and sports analytics. He has served for over 45 years on the editorial boards of Operations Research and Management Science, and given over 400 academic presentations, 200 invited presentations in 28 countries, 20 conference keynote presentations, and 10 INFORMS tutorials. A 2008 citation study ranked him 13th among 1,376 operations management scholars. In 2018, he was President of INFORMS. He is a Fellow of INFORMS.

In 2022, he published *Supply Chain Scheduling* (Springer, 706 pp), with Zhi-Long Chen.

Siyi Wang is a PhD student of the School of Science and Engineering, The Chinese University of Hong Kong, Shenzhen, Guangdong. She holds a bachelor's degree in engineering and a bachelor's degree in economics from University of Electronic Science and Technology of China, Chengdu, Sichuan. Her research interests are in supply chain management, and information asymmetry. Her works have been published in *Transportation Research Part E: Logistics and Transportation Review*, *Chinese Journal of Management Science*, and *Systems Engineering-Theory & Practice*.

Feng Zhang earned his PhD degree in engineering management from The Chinese University of Hong Kong. He is now the risk director in Suning Bank (the fifth among the world's top 20 emerging banks, as rated by Moody's in 2022), Nanjing, Jiangsu. His major responsibilities include enterprise risk management, credit risk measurement, analysis, evaluation, and early-warning and treatment. He used to working in the Agriculture Bank of China for over ten years, during which he reviewed and approved over 1350 projects with amount of more than RMB 55 billion. He was granted the title of Senior Economist in 2019.